OPENING PROBLEM

A **plane** is a flat surface that extends forever, and has zero thickness.

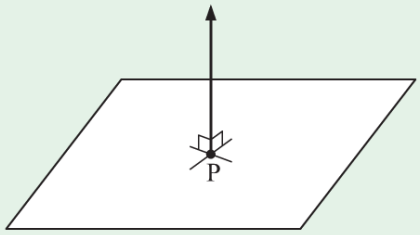
Things to think about:

a Given that a plane is 2-dimensional, does it make sense to talk about a plane having *direction*?

b Suppose you are at a point P on a plane.

i In how many different directions could you move from P and still be on the plane?

ii Is there a direction which is perpendicular to *all* of the directions you considered in **i**? If so, how would you find that direction?



c Can we use a point and a direction vector to specify a *plane*?

d If there are two planes in space, do they necessarily meet?

There are many applications of vectors in geometry. Some of these applications we do not use in 2-dimensional planar geometry because we have other tools.

In 3-dimensional space, however, vector methods are very efficient, in particular when we consider the relationships between lines and planes.

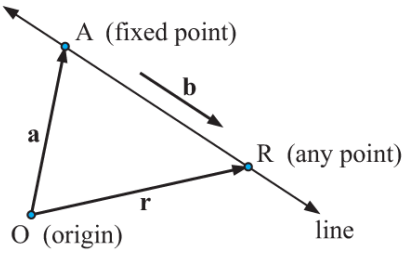
A LINES IN 2 AND 3 DIMENSIONS

In both 2-dimensional and 3-dimensional geometry we can determine the **equation of a line** using its **direction** and any **fixed point** on the line.

In 2 dimensions we are dealing with a **line in a plane**. In 3 dimensions we are dealing with a **line in space**.

Suppose a line passes through a fixed point A with position vector **a**, and the line is parallel to the vector **b**.

Consider a point R on the line with position vector **r**.





By vector addition, $\mathbf{r} = \mathbf{a} + \overrightarrow{\text{AR}}$.

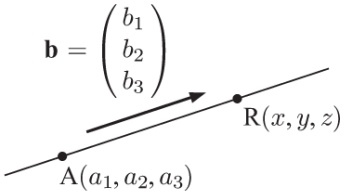
Since $\overrightarrow{\text{AR}}$ is parallel to **b**,

$$\overrightarrow{\text{AR}} = \lambda\mathbf{b} \quad \text{for some scalar } \lambda \in \mathbb{R}$$
$$\therefore \quad \mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$$

So, $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, $\lambda \in \mathbb{R}$ is the **vector equation** of the line.

Suppose $\text{R}(x, y, z)$ is any point on the line,
$\text{A}(a_1, a_2, a_3)$ is the known or fixed point on the line,

and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ is the **direction vector** of the line.



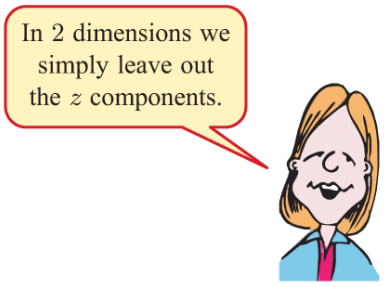
- $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \lambda \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ is the **vector equation** of the line.

- The **parametric equations** of the line are:

$$\begin{cases} x = a_1 + \lambda b_1 \\ y = a_2 + \lambda b_2 \\ z = a_3 + \lambda b_3 \end{cases} \quad \text{where } \lambda \in \mathbb{R} \text{ is the } \textbf{parameter}.$$

- By equating λ values, we obtain the **Cartesian equation** of the line $\dfrac{x - a_1}{b_1} = \dfrac{y - a_2}{b_2} = \dfrac{z - a_3}{b_3}$.

  In 2 dimensions, the Cartesian form can be written in the familiar form $ax + by = d$.



DISCUSSION

In 2 dimensions we talk about the gradient of a line.

- If a line in 2 dimensions has direction vector $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, what is its gradient?
- Does it make sense to talk about the gradient of a line in space?

Example 1

Self Tutor

A line in space passes through $(1, -2, 3)$ in the direction $4\mathbf{i} + 5\mathbf{j} - 6\mathbf{k}$.

Describe the line using:

a a vector equation **b** parametric equations **c** a Cartesian equation.

a The vector equation is $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ -6 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

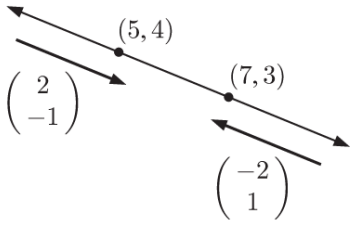
b The parametric equations are: $x = 1 + 4\lambda, \quad y = -2 + 5\lambda, \quad z = 3 - 6\lambda, \quad \lambda \in \mathbb{R}$.

c Equating λ values, the Cartesian equations are: $\dfrac{x-1}{4} = \dfrac{y+2}{5} = \dfrac{z-3}{-6}$.

NON-UNIQUENESS OF THE VECTOR EQUATION OF A LINE

Consider the line passing through $(5, 4)$ and $(7, 3)$. When writing the equation of the line, we could use either point to give the position vector $\mathbf{a}$.

Similarly, we could use the direction vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$, but we could also use $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ or any non-zero scalar multiple of these vectors.



We could thus write the equation of the line as

$$\mathbf{r} = \begin{pmatrix} 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \ \lambda \in \mathbb{R} \quad \text{or} \quad \mathbf{r} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \ \mu \in \mathbb{R} \quad \text{and so on.}$$

Notice how we use different parameters λ and μ when we write these equations. This is because the parameters are clearly not the same: when $\lambda = 0$, we have the point $(5, 4)$
when $\mu = 0$, we have the point $(7, 3)$.

In fact, the parameters are related by $\mu = 1 - \lambda$.

Example 2 **Self Tutor**

Find parametric equations for the line through $A(2, -1, 4)$ and $B(-1, 0, 2)$.

We require a direction vector for the line, either $\overrightarrow{AB}$ or $\overrightarrow{BA}$.

$$\overrightarrow{AB} = \begin{pmatrix} -1-2 \\ 0-(-1) \\ 2-4 \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \\ -2 \end{pmatrix}$$

Using point A, the equations are:

$$x = 2 - 3\lambda, \quad y = -1 + \lambda, \quad z = 4 - 2\lambda, \quad \lambda \in \mathbb{R}$$

or using point B, the equations are:

$$x = -1 - 3\mu, \quad y = \mu, \quad z = 2 - 2\mu, \quad \mu \in \mathbb{R}.$$

EXERCISE 13A

1 Describe each of the following lines using:

i a vector equation **ii** parametric equations **iii** a Cartesian equation.

a a line with direction $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ which passes through $(1, 4)$

b a line passing through $(5, 2)$ which is perpendicular to $\begin{pmatrix} 5 \\ 2 \end{pmatrix}$

c a line parallel to $3\mathbf{i} + 7\mathbf{j}$ which cuts the x-axis at -6

d a line perpendicular to $3\mathbf{i} - \mathbf{j}$ which cuts the y-axis at 2

e a line passing through $(3, 0)$ and $(7, 2)$

f a line passing through $(-2, 5)$ and $(4, -6)$.

2 A line passes through $(4, -3)$ with direction vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

a Write parametric equations for the line using the parameter λ.

b Find the points on the line for which $\lambda = 0, 1, 2, -1$, and -3.

3 Line L has vector equation $\mathbf{r} = \begin{pmatrix} 1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 3 \end{pmatrix}, \ \lambda \in \mathbb{R}$.

a Locate the point on the line corresponding to $\lambda = 1$.

b Explain why the direction of the line could also be described by $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$.

c Use your answers to **a** and **b** to write an alternative vector equation for line L.

4 Describe each of the following lines using:

i a vector equation **ii** parametric equations **iii** a Cartesian equation.

a a line parallel to $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ which passes through $(1, 3, -7)$

b a line which passes through $(0, 1, 2)$ with direction vector $\mathbf{i} + \mathbf{j} - 2\mathbf{k}$

c a line parallel to the X-axis which passes through $(-2, 2, 1)$

d a line parallel to $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ which passes through $(0, 2, -1)$

e a line perpendicular to the XY-plane which passes through $(3, 2, -1)$.

5 Find the vector equation of the line which passes through:

a A$(1, 2, 1)$ and B$(-1, 3, 2)$ **b** C$(0, 1, 3)$ and D$(3, 1, -1)$

c E$(1, 2, 5)$ and F$(1, -1, 5)$ **d** G$(0, 1, -1)$ and H$(5, -1, 3)$.

6 Find the direction vector of the line:

a $\begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 0 \\ 3 \end{pmatrix}, \quad \lambda \in \mathbb{R}$ **b** $x = 5 - t, \quad y = 1 + t, \quad z = 2 - 3t, \quad t \in \mathbb{R}$

c $\dfrac{x-2}{3} = \dfrac{y+1}{2} = z - 1$ **d** $\dfrac{1-x}{2} = \dfrac{y}{4} = \dfrac{z-3}{3}$

7 Find the coordinates of the point where the line with parametric equations $x = 1 - \lambda$, $y = 3 + \lambda$, and $z = 3 - 2\lambda$, $\lambda \in \mathbb{R}$ meets:

a the YZ-plane $x = 0$ **b** the XZ-plane $y = 0$ **c** the XY-plane $z = 0$.

8 The parametric equations of a line are $x = x_0 + \lambda l$, $y = y_0 + \lambda m$, $z = z_0 + \lambda n$ for $\lambda \in \mathbb{R}$.

a Find the coordinates of the point on the line corresponding to $\lambda = 0$.

b Find the direction vector of the line.

c Write the Cartesian equation for the line.

9 Find points on the line with parametric equations $x = 2 - \lambda$, $y = 3 + 2\lambda$, and $z = 1 + \lambda$, $\lambda \in \mathbb{R}$ which are $5\sqrt{3}$ units from the point $(1, 0, -2)$.

10 A line has parametric equations $x = 1 + \lambda$, $y = 2 - \lambda$, $z = 3 + \lambda$.

a Find the distance from P$(1, 1, 2)$ to a general point X on the line.

b Use quadratic theory to find the value of λ which minimises the distance PX.

c Show that when the distance PX is minimised, $\overrightarrow{\text{PX}}$ is perpendicular to the line.

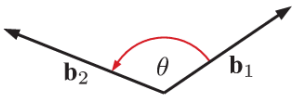
11 Line L_1 has equation $\mathbf{r} = \begin{pmatrix} -3 \\ 1 \\ 4 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$. Line L_2 passes through $(2, 0, 3)$ and $(-2, 1, 3)$.

a Find the equation of L_2.

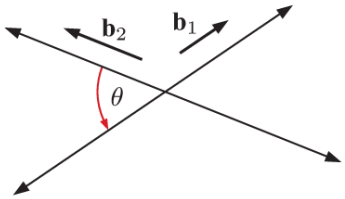
b The line L_3 is perpendicular to both L_1 and L_2, and cuts the Z-axis at -2. Find the equation of L_3 in Cartesian form.

B THE ANGLE BETWEEN TWO LINES

In **Chapter 12** we saw that the angle between two vectors is measured in the range $0^\circ \leqslant \theta \leqslant 180^\circ$, and found using the formula $\cos\theta = \dfrac{\mathbf{b}_1 \bullet \mathbf{b}_2}{|\mathbf{b}_1||\mathbf{b}_2|}$.



Since lines continue infinitely in both directions, we talk about the *acute* angle between them. For an acute angle, $\cos\theta > 0$, so we use the formula



$$\cos\theta = \frac{|\mathbf{b}_1 \bullet \mathbf{b}_2|}{|\mathbf{b}_1||\mathbf{b}_2|}$$

where $\mathbf{b}_1$ and $\mathbf{b}_2$ are the direction vectors of the given lines.

Example 3 Self Tutor

Find the angle between the lines L_1: $x = 2 - 3t$, $y = -1 + t$ and L_2: $x = 1 + 2s$, $y = -4 + 3s$, where $s, t \in \mathbb{R}$

L_1 has direction vector $\mathbf{b}_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$. L_2 has direction vector $\mathbf{b}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

Now $\cos\theta = \dfrac{|\mathbf{b}_1 \bullet \mathbf{b}_2|}{|\mathbf{b}_1||\mathbf{b}_2|} = \dfrac{|-6+3|}{\sqrt{10}\sqrt{13}} \approx 0.2631$

$\therefore \ \theta \approx 74.7^\circ$

$\therefore$ the angle between L_1 and L_2 is about 74.7°.

PARALLEL AND PERPENDICULAR LINES

Consider two lines $\mathbf{r}_1 = \mathbf{a}_1 + \lambda\mathbf{b}_1$ and $\mathbf{r}_2 = \mathbf{a}_2 + \mu\mathbf{b}_2$.

- $\mathbf{r}_1$ and $\mathbf{r}_2$ are **parallel** if $\mathbf{b}_1 = k\mathbf{b}_2$ for some scalar k. We write $\mathbf{r}_1 \parallel \mathbf{r}_2$.
- $\mathbf{r}_1$ and $\mathbf{r}_2$ are **perpendicular** if $\mathbf{b}_1 \bullet \mathbf{b}_2 = 0$. We write $\mathbf{r}_1 \perp \mathbf{r}_2$.

Example 4 Self Tutor

Find the angle between the lines:

$$L_1: \ \mathbf{r}_1 = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 1 \\ -2 \end{pmatrix}, \ \lambda \in \mathbb{R} \quad \text{and} \quad L_2: \ \frac{1-x}{3} = y = \frac{2-z}{2}.$$

L_1 has direction vector $\mathbf{b}_1 = \begin{pmatrix} -3 \\ 1 \\ -2 \end{pmatrix}$. L_2 has direction vector $\mathbf{b}_2 = \begin{pmatrix} -3 \\ 1 \\ -2 \end{pmatrix}$.

Since $\mathbf{b}_1 = \mathbf{b}_2$, the lines are parallel so the angle between them is 0°.

EXERCISE 13B

1 Find the angle between the lines L_1: $x = -4 + 12t$, $y = 3 + 5t$, $t \in \mathbb{R}$
and L_2: $x = 3s$, $y = -6 - 4s$, $s \in \mathbb{R}$.

2 Show that the lines L_1: $x = 2 + 5p$, $y = 19 - 3p$, $z = 9 + 2p$, $p \in \mathbb{R}$
and L_2: $x = 3 + 4r$, $y = 7 + 10r$, $z = -3 + 5r$, $r \in \mathbb{R}$ are perpendicular.

3 The line L_1 passes through $(-6, 3)$ and is parallel to $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$.

The line L_2 has direction $5\mathbf{i} + 4\mathbf{j}$ and cuts the y-axis at 8.

Find the angle between the lines.

4 **a** Find the angle between the lines:

$$L_1: \ \frac{x-8}{3} = \frac{9-y}{16} = \frac{z-10}{7} \qquad \text{and} \qquad L_2: \ \mathbf{r}_2 = \begin{pmatrix} 15 \\ 29 \\ 5 \end{pmatrix} + t\begin{pmatrix} 3 \\ 8 \\ -5 \end{pmatrix}, \ t \in \mathbb{R}.$$

b A third line L_3 is perpendicular to L_1 and has direction vector $-3\mathbf{j} + a\mathbf{k}$.

i Find the value of a. **ii** Find the angle between L_2 and L_3.

5 The line L_1 passes through $(-1, 5, 2)$ and $(0, 2, -2)$, and the line L_2 is parallel to $\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$. Find the angle between the lines.

6 The line L_1 has equation $\dfrac{x-2}{2} = \dfrac{y+3}{3} = 1 - z$. The line L_2 meets L_1 on the X-axis and passes through $(5, -3, 1)$. Find the angle between the lines.

Example 5 **Self Tutor**

Find the measure of the angle between the lines $2x + y = 5$ and $3x - 2y = 8$.

$2x + y = 5$ has gradient $-\frac{2}{1}$ $\therefore$ it has direction vector $\mathbf{b}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

$3x - 2y = 8$ has gradient $\frac{3}{2}$ $\therefore$ it has direction vector $\mathbf{b}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

If a line has gradient $\frac{b}{a}$, it has direction vector $\begin{pmatrix} a \\ b \end{pmatrix}$.



If the angle between the lines is θ, then

$$\cos\theta = \frac{|\mathbf{b}_1 \bullet \mathbf{b}_2|}{|\mathbf{b}_1||\mathbf{b}_2|} = \frac{|(1 \times 2) + (-2 \times 3)|}{\sqrt{1+4}\sqrt{4+9}} = \frac{4}{\sqrt{5}\sqrt{13}}$$

$$\therefore \ \theta = \cos^{-1}\left(\tfrac{4}{\sqrt{65}}\right) \approx 60.3°$$

$\therefore$ the angle between the lines is about $60.3°$.

7 Find the measure of the angle between the lines:

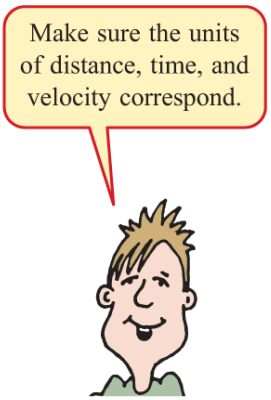
a $x - y = 3$ and $3x + 2y = 11$ **b** $y = x + 2$ and $y = 1 - 3x$

c $y + x = 7$ and $x - 3y + 2 = 0$ **d** $y = 2 - x$ and $x - 2y = 7$

C CONSTANT VELOCITY PROBLEMS

An object moving with a constant velocity will travel in a straight line. To model the position using vectors:

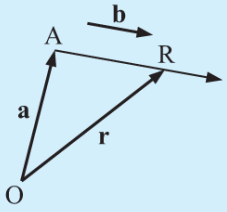
- the **velocity vector** of the motion gives the direction vector of the line
- **time** is the parameter
- the **initial position** of the object gives a fixed point on the line.



If an object has initial position vector **a** and moves with constant velocity **b**, its position at time t is given by

$$\mathbf{r} = \mathbf{a} + t\mathbf{b} \quad \text{for} \quad t \geqslant 0.$$

The **speed** of the object is $|\mathbf{b}|$.



Example 6 Self Tutor

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 9 \end{pmatrix} + t\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ is the vector equation of the path of an object.

The time t is in seconds, $t \geqslant 0$. The distance units are metres.



a Find the object's initial position.

b Plot the path of the object for $t = 0, 1, 2, 3$.

c Find the velocity vector of the object.

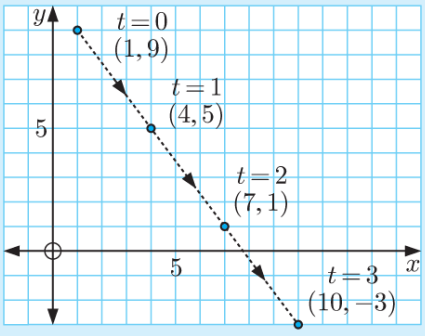
d Find the object's speed.

e If the object continues in the same direction but increases its speed to 30 m s^{-1}, state its new velocity vector.

a At $t = 0$, $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 9 \end{pmatrix}$

$\therefore$ the object is at $(1, 9)$.

b



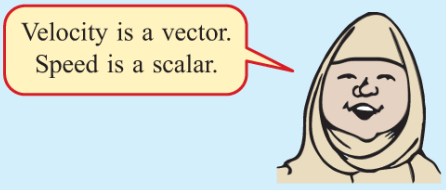
c The velocity vector is $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.

d The speed is $\left|\begin{pmatrix} 3 \\ -4 \end{pmatrix}\right| = \sqrt{3^2 + (-4)^2}$

$= 5$ m s^{-1}.

e Previously, the speed was 5 m s^{-1} and the velocity vector was $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.

$\therefore$ the new velocity vector is $\frac{30}{5}\begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 18 \\ -24 \end{pmatrix}$.



Example 7 **Self Tutor**

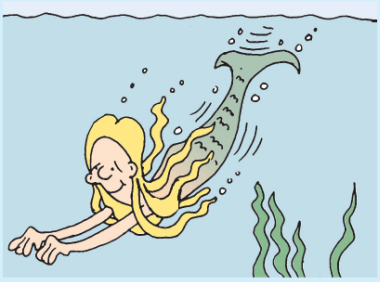
A mermaid is initially at $(15, 8, -1)$. She swims with velocity vector $3\mathbf{i} - 5\mathbf{j} - \frac{9}{2}\mathbf{k}$ metres per second.

Find:

a the position of the mermaid after t seconds

b the speed of the mermaid

c the time when the mermaid reaches depth 28 m.



a $\mathbf{r} = \mathbf{a} + t\mathbf{b}$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 15 \\ 8 \\ -1 \end{pmatrix} + t\begin{pmatrix} 3 \\ -5 \\ -\frac{9}{2} \end{pmatrix} = \begin{pmatrix} 15+3t \\ 8-5t \\ -1-\frac{9}{2}t \end{pmatrix}, \quad t \in \mathbb{R}$$

After t seconds the mermaid is at $(15+3t,\ 8-5t,\ -1-\frac{9}{2}t)$.

b The speed of the mermaid

$= \sqrt{3^2 + (-5)^2 + (-\frac{9}{2})^2}$

$= \sqrt{54\frac{1}{4}}$

$\approx 7.37 \text{ m s}^{-1}$

c The mermaid reaches depth 28 m when

$z(t) = -1 - \frac{9}{2}t = -28$

$\therefore \ \frac{9}{2}t = 27$

$\therefore \ t = 6$ seconds

EXERCISE 13C

1 A particle at $P(x(t), y(t))$ moves such that $x(t) = 1 + 2t$ and $y(t) = 2 - 5t$, $t \geqslant 0$. The distances are in centimetres and t is in seconds.

a Find the initial position of P.

b Plot the path of P for $t = 0, 1, 2, 3$.

c Find the velocity vector of P.

d Find the speed of P.

e A second particle is travelling in the opposite direction, but with speed 8 cm s^{-1}. Find its velocity vector.

2 Each of the following vector equations represents the path of a moving object. t is measured in seconds, $t \geqslant 0$. Distances are measured in metres. In each case, find:

i the initial position **ii** the velocity vector **iii** the speed of the object.

a $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \\ 0 \end{pmatrix} + t\begin{pmatrix} 12 \\ 5 \\ 6 \end{pmatrix}$

b $x = 3 + 2t, \ y = -t, \ z = 4 - 2t$

3 **a** Find the vector equation of a boat initially at $(2, 3)$, which travels with velocity vector $\begin{pmatrix} 4 \\ -5 \end{pmatrix}$. The grid units are kilometres, and the time is in hours.

b Locate the boat's position after 90 minutes.

c How long will it take for the boat to reach the point $(5, -0.75)$?

4 A remote controlled toy car is initially at the point $(-3, -2)$. It moves with constant velocity $2\mathbf{i} + 4\mathbf{j}$. The distance units are centimetres, and the time is in seconds.

a Write an expression for the position vector of the car at any time $t \geqslant 0$.

b Hence find the position vector of the car at time $t = 2.5$.

c Find when the car is **i** due north **ii** due west of the observation point $(0, 0)$.

d Plot the car's position at times $t = 0, 0.5, 1, 1.5, 2, 2.5$.

5 Find the velocity vector of:

a a speed boat moving parallel to $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$ with speed 150 km h^{-1}

b a jogger moving parallel to $\begin{pmatrix} -5 \\ 12 \end{pmatrix}$ with speed 7.8 km h^{-1}

c a ferry moving parallel to $6\mathbf{i} + 7\mathbf{j}$ with speed 25 km h^{-1}

d a hot air balloon moving parallel to $4\mathbf{i} + 7\mathbf{j} + \mathbf{k}$ with speed 33 km h^{-1}

e a swooping eagle moving in the direction $-2\mathbf{i} + 5\mathbf{j} - 14\mathbf{k}$ with speed 90 km h^{-1}.

6 Yacht A moves according to $x_A(t) = 4 + t$, $y_A(t) = 5 - 2t$, $t \geqslant 0$. Yacht B moves according to $x_B(t) = 1 + 2t$, $y_B(t) = -8 + t$, $t \geqslant 0$. The distance units are kilometres, and the time units are hours.

a Find the initial position of each yacht. **b** Find the velocity vector of each yacht.

c Show that the speed of each yacht is constant, and state these speeds.

d Show, using vector algebra, that the paths of the yachts are at right angles to each other.

7 At 1:34 pm, submarine P fires a torpedo with velocity vector $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ from the point $(-5, 4)$. a minutes later submarine Q fires a torpedo with velocity vector $\begin{pmatrix} -4 \\ -3 \end{pmatrix}$ from the point $(15, 7)$. Distances are measured in kilometres, and time is in minutes.

a Show that the position of P's torpedo can be written as $P(x_1(t), y_1(t))$ where $x_1(t) = -5+3t$ and $y_1(t) = 4 - t$.

b What is the speed of P's torpedo?

c Show that the position of Q's torpedo can be written as $Q(x_2(t), y_2(t))$ where $x_2(t) = 15 - 4(t - a)$ and $y_2(t) = 7 - 3(t - a)$.

d Find the angle between the paths of the torpedoes.

e Q's torpedo is successful in knocking out P's torpedo. At what time did Q fire its torpedo and at what time did the explosion occur?

8 A cable car leaves from $(10, 3, 0)$. It moves in the direction $2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ at 4.5 m s^{-1}.

a Find the velocity vector of the cable car.

b Find the position of the cable car after 30 seconds.

c The destination of the cable car has X-coordinate 550. Find the length of the cable car ride.

d At what angle to the horizontal does the cable car travel?



9 A helicopter at $A(6, 9, 3)$ moves with constant velocity. 10 minutes later it is at $B(3, 10, 2.5)$. Distances are in kilometres.

a Find $\overrightarrow{AB}$.

b Find the helicopter's speed.

c Determine the vector equation which represents the path of the helicopter.

d The helicopter is travelling directly towards its helipad, which has Z-coordinate 0. Find the total time taken for the helicopter to land.

e At what angle to the horizontal is the helicopter flying?

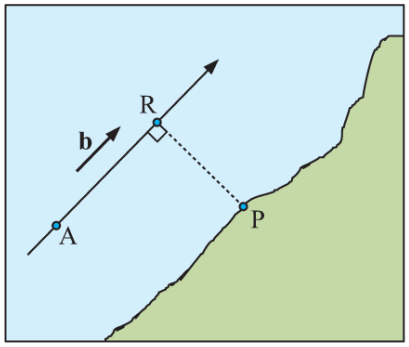
D THE SHORTEST DISTANCE FROM A POINT TO A LINE

A ship R sails through point A in the direction **b** and continues past a port P.

The ship is *closest* to the port when [PR] is perpendicular to [AR].

$\therefore \quad \overrightarrow{PR} \bullet \mathbf{b} = 0$

In this situation, point R is called the **foot of the perpendicular from P to the line**.



Example 8 Self Tutor

A line has vector equation $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + t\begin{pmatrix} 3 \\ -1 \end{pmatrix}$, $t \in \mathbb{R}$. Find the shortest distance from the point $P(5, -1)$ to the line.

Let N be the point on the line closest to P.

N has coordinates $(1 + 3t, 2 - t)$ for some t, and $\overrightarrow{PN}$ is $\begin{pmatrix} 1 + 3t - 5 \\ 2 - t - (-1) \end{pmatrix} = \begin{pmatrix} 3t - 4 \\ 3 - t \end{pmatrix}$.

The distance between P and the line is minimised when $\overrightarrow{PN} \bullet \begin{pmatrix} 3 \\ -1 \end{pmatrix} = 0$.

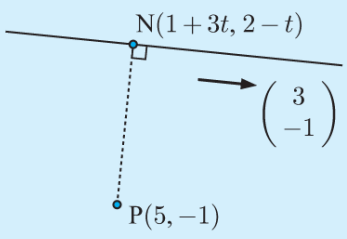
$\therefore \quad \begin{pmatrix} 3t - 4 \\ 3 - t \end{pmatrix} \bullet \begin{pmatrix} 3 \\ -1 \end{pmatrix} = 0$

$\therefore \quad 3(3t - 4) - (3 - t) = 0$

$\therefore \quad 9t - 12 - 3 + t = 0$

$\therefore \quad 10t = 15$

$\therefore \quad t = \frac{15}{10} = \frac{3}{2}$



Thus $\overrightarrow{PN} = \begin{pmatrix} \frac{9}{2} - 4 \\ 3 - \frac{3}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{3}{2} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$\therefore \quad \overrightarrow{PN} = \frac{1}{2}\sqrt{1^2 + 3^2} = \frac{\sqrt{10}}{2}$ units

$\therefore$ the shortest distance from the point P to the line is $\frac{\sqrt{10}}{2}$ units.

EXERCISE 13D

1 Find the shortest distance from:

a $P(3, 2)$ to the line with parametric equations $x = 2 + t, \ y = 3 + 2t, \ t \in \mathbb{R}$

b $Q(-1, 1)$ to the line with parametric equations $x = t, \ y = 1 - t, \ t \in \mathbb{R}$

c $R(-3, -1)$ to the line $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + s\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \ s \in \mathbb{R}$

d $S(5, -2)$ to the line $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix} + t\begin{pmatrix} 3 \\ -7 \end{pmatrix}, \ t \in \mathbb{R}$.

2 An ocean liner is at $(6, -6)$, cruising at 10 km h^{-1} in the direction $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$.
A fishing boat is anchored at $(0, 0)$. Distances are in kilometres.

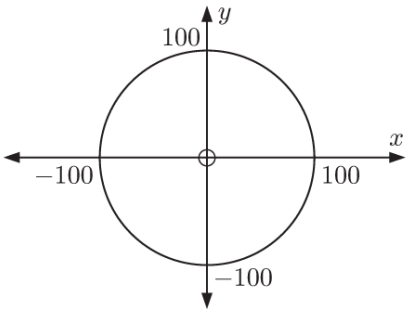
a Find, in terms of **i** and **j**, the initial position vector of the liner from the fishing boat.

b Write an expression for the position vector of the liner at any time t hours after it has sailed from $(6, -6)$.

c When will the liner be due east of the fishing boat?

d Find the time and position of the liner when it is nearest to the fishing boat.

3 Let **i** represent a displacement 1 km due east
and **j** represent a displacement 1 km due north.
The control tower of an airport is at $(0, 0)$. Aircraft within 100 km of $(0, 0)$ are visible on the radar screen at the control tower.
At 12:00 noon an aircraft is 200 km east and 100 km north of the control tower. It is flying parallel to the vector $\mathbf{b} = -3\mathbf{i} - \mathbf{j}$ with speed $40\sqrt{10}$ km h^{-1}.



a Write down the velocity vector of the aircraft.

b Write a vector equation for the path of the aircraft using t to represent the time in hours that has elapsed since 12:00 noon.

c Find the position of the aircraft at 1:00 pm.

d Show that the aircraft first becomes visible on the radar screen at 1:00 pm.

e Find the time when the aircraft is closest to the control tower, and find the distance between the aircraft and the control tower at this time.

f At what time will the aircraft disappear from the radar screen?

Example 9 Self Tutor

Consider the point $P(-1, 2, 3)$ and the line with parametric equations $x = 1 + 2t, \ y = -4 + 3t, \ z = 3 + t, \ t \in \mathbb{R}$.

a Find the coordinates of the foot of the perpendicular from P to the line.

b Find the shortest distance from the point to the line.

a The line has direction vector $\mathbf{b} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$.

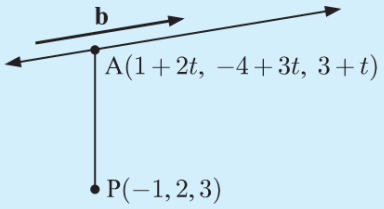
Let $A(1+2t,\ -4+3t,\ 3+t)$ be any point on the given line.

$$\therefore\ \overrightarrow{PA} = \begin{pmatrix} 1+2t-(-1) \\ -4+3t-2 \\ 3+t-3 \end{pmatrix} = \begin{pmatrix} 2+2t \\ -6+3t \\ t \end{pmatrix}$$

If A is the closest point on the line to P, then $\overrightarrow{PA} \perp \mathbf{b}$.

$$\therefore\ \overrightarrow{PA} \bullet \mathbf{b} = 0$$

$$\therefore\ \begin{pmatrix} 2+2t \\ -6+3t \\ t \end{pmatrix} \bullet \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = 0$$



$$\therefore\ 2(2+2t)+3(-6+3t)+1(t) = 0$$

$$\therefore\ 4+4t-18+9t+t = 0$$

$$\therefore\ 14t = 14$$

$$\therefore\ t = 1$$

Substituting $t = 1$ into the parametric equations, the foot of the perpendicular is $(3, -1, 4)$.

b Using $t = 1$, $\overrightarrow{PA} = \begin{pmatrix} 2+2 \\ -6+3 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \\ 1 \end{pmatrix}$

$$\therefore\ |\overrightarrow{PA}| = \sqrt{4^2+(-3)^2+1^2} = \sqrt{26} \text{ units}$$

So, the shortest distance from P to the line is $\sqrt{26}$ units.

4 **a** Find the coordinates of the foot of the perpendicular from $(3, 0, -1)$ to the line with parametric equations $x = 2+3t$, $y = -1+2t$, $z = 4+t$, $t \in \mathbb{R}$.

b Hence find the shortest distance from the line to the point.

5 **a** Find the coordinates of the foot of the perpendicular from $(1, 1, 3)$ to the line with vector equation $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + t\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$, $t \in \mathbb{R}$.

b Hence find the shortest distance from the point to the line.

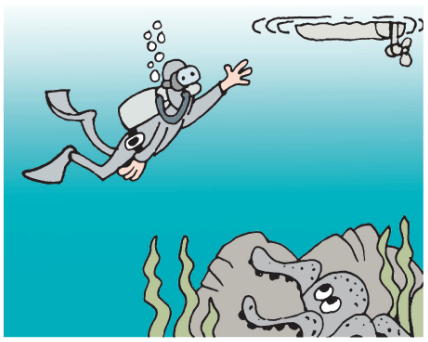
6 A diver swims from $(12, 25, -20)$ back to his boat at $(7, -15, 0)$, at a speed of $0.9\ \text{m s}^{-1}$.

a Find the velocity vector of the diver.

b An octopus watches the diver from its home at $(12, -8, -5)$.

i At what time is the diver closest to the octopus?

ii Find the shortest distance from the diver to the octopus.



E INTERSECTING LINES

Vector equations of two intersecting lines can be **solved simultaneously** to find the point where the lines meet.

Example 10 Self Tutor

Line 1 has vector equation $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} + s\begin{pmatrix} 3 \\ 2 \end{pmatrix}$, $s \in \mathbb{R}$.

Line 2 has vector equation $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 15 \\ 5 \end{pmatrix} + t\begin{pmatrix} -4 \\ 1 \end{pmatrix}$, $t \in \mathbb{R}$.

Find the point where the two lines meet.

The lines meet where $\begin{pmatrix} -2 \\ 1 \end{pmatrix} + s\begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 15 \\ 5 \end{pmatrix} + t\begin{pmatrix} -4 \\ 1 \end{pmatrix}$

$\therefore\ -2 + 3s = 15 - 4t$ and $1 + 2s = 5 + t$

$\therefore\ 3s + 4t = 17$ .... (1) and $2s - t = 4$ .... (2)

$3s + 4t = 17$ $\{(1)\}$

$8s - 4t = 16$ $\{(2) \times 4\}$

$\therefore\ 11s = 33$

$\therefore\ s = 3$

This method involves solving simultaneous equations by **elimination**.



Using (2), $2(3) - t = 4$

$\therefore\ t = 2$

Using line 1, $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} + 3\begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ 7 \end{pmatrix}$

Checking in line 2, $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 15 \\ 5 \end{pmatrix} + 2\begin{pmatrix} -4 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 7 \end{pmatrix}$ ✓

$\therefore$ the lines meet at $(7, 7)$.

EXERCISE 13E

1 Triangle ABC is formed by three lines:

(AB) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 6 \end{pmatrix} + r\begin{pmatrix} 3 \\ -2 \end{pmatrix}$, (AC) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + s\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and

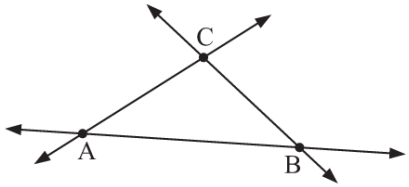
(BC) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10 \\ -3 \end{pmatrix} + t\begin{pmatrix} -2 \\ 3 \end{pmatrix}$, where r, s, and t are scalars.

a Draw the three lines accurately on a grid.

b Hence find the coordinates of A, B, and C.

c Prove that triangle ABC is isosceles.

d Use vector methods to check your answers to **b**.



2 The parallelogram ABCD is defined by four lines:

(AB) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \end{pmatrix} + r\begin{pmatrix} 7 \\ 3 \end{pmatrix}$, (AD) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \end{pmatrix} + s\begin{pmatrix} 1 \\ 2 \end{pmatrix}$,

(CD) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 22 \\ 25 \end{pmatrix} + t\begin{pmatrix} -7 \\ -3 \end{pmatrix}$, (CB) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 22 \\ 25 \end{pmatrix} + u\begin{pmatrix} -1 \\ -2 \end{pmatrix}$,

where r, s, t, and u are scalars.

a Draw an accurate sketch of the four lines and the parallelogram formed by them.

b Hence find the coordinates of A, B, C, and D.

c Use vector methods to check your answers to **b**.

3 The isosceles triangle ABC is defined by three lines:

(AB) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + r\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, (BC) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \end{pmatrix} + s\begin{pmatrix} -1 \\ -2 \end{pmatrix}$, and

(AC) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + t\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, where r, s and t are scalars.

a Use vector methods to find the coordinates of A, B, and C.

b Find all side lengths of the triangle, and hence determine which two sides are equal in length.

4 (QP) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} + r\begin{pmatrix} 14 \\ 10 \end{pmatrix}$, (QR) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} + s\begin{pmatrix} 17 \\ -9 \end{pmatrix}$, and

(PR) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 18 \end{pmatrix} + t\begin{pmatrix} 5 \\ -7 \end{pmatrix}$, where r, s, and t are scalars.

a Use vector methods to find the coordinates of P, Q, and R.

b Find vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ and evaluate $\overrightarrow{PQ} \bullet \overrightarrow{PR}$.

c Hence find the size of $\widehat{QPR}$.

d Find the area of triangle PQR.

5 (AB) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix} + r\begin{pmatrix} 4 \\ 1 \end{pmatrix}$, (BC) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 18 \\ 9 \end{pmatrix} + s\begin{pmatrix} -8 \\ 32 \end{pmatrix}$,

(CD) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 14 \\ 25 \end{pmatrix} + t\begin{pmatrix} -8 \\ -2 \end{pmatrix}$, and (AD) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + u\begin{pmatrix} -3 \\ 12 \end{pmatrix}$,

where r, s, t, and u are scalars.

a Use vector methods to find the coordinates of A, B, C, and D.

b Find:

i $\overrightarrow{AC}$ and $\overrightarrow{DB}$ **ii** $|\overrightarrow{AC}|$ and $|\overrightarrow{DB}|$ **iii** $\overrightarrow{AC} \bullet \overrightarrow{DB}$

iv the midpoints of [AC] and [DB].

c What do the answers to **b** tell you about quadrilateral ABCD?

THEORY OF KNOWLEDGE

In his 1827 book entitled *The Barycentric Calculus*, German mathematician **August Möbius** demonstrated the use of directed line segments in projective geometry. He discussed the addition and scalar multiplication of directed line segments, but he did not name them vectors.

Gauss, **Grassmann**, **Laplace**, and **Lagrange** all contributed to build the field we know as vector analysis today.

An important contribution was made in 1843 by **Sir William Hamilton**, who defined a *quaternion* as the quotient of two directed line segments.

The American mathematician **Josiah Willard Gibbs** (1839 - 1903) began teaching a course on vector analysis at Yale University in 1879. He published *Elements of Vector Analysis* in two halves, and in 1901 his work was summarised by his student **Edwin Wilson** in the book *Vector Analysis*.

Oliver Heaviside (1850 - 1925) developed vector analysis separately from Gibbs, receiving a copy of Gibbs' work later, in 1888. Heaviside wrote:



Sir William Hamilton

"... the invention of quaternions must be regarded as a most remarkable feat of human ingenuity. Vector analysis, without quaternions, could have been found by any mathematician by carefully examining the mechanics of the Cartesian mathematics; but to find out quaternions required a genius."

1. Are geometry and algebra two separate domains of knowledge?
2. Given that Gibbs and Heaviside developed vector analysis independently, who can claim to be the founder of this field?
3. Does mathematics *evolve*?

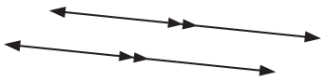
F RELATIONSHIPS BETWEEN LINES

We have just seen how the intersection of two lines in parametric form can be found by vector methods. We have also seen how to determine whether two lines are parallel.

We now consider finding the possible relationships between lines in Cartesian form. We will consider lines in both 2 and 3 dimensions.

LINE CLASSIFICATION IN 2 DIMENSIONS

There are three possible relationships between two lines in a plane:

intersecting	**parallel**	**coincident**
		
one point of intersection unique solution	lines do not meet no solutions	the same line infinitely many solutions

In order to classify a pair of lines in 2 dimensions, we attempt to solve the equations of the lines simultaneously using row reduction.

Example 11 Self Tutor

Reduce the system $\begin{cases} 3x + 5y = 9 \\ 2x - 3y = -13 \end{cases}$ using row reduction. Hence describe the relationship between the pair of lines.

In augmented matrix form, the system is

$$\left(\begin{array}{cc|c} 3 & 5 & 9 \\ 2 & -3 & -13 \end{array}\right)$$

$$\sim \left(\begin{array}{cc|c} 1 & \frac{5}{3} & 3 \\ 0 & 19 & 57 \end{array}\right) \quad \begin{array}{l} \frac{1}{3}R_1 \to R_2 \\ 2R_1 - 3R_2 \to R_2 \end{array} \longleftarrow \left\{\begin{array}{ccc} 6 & 10 & 18 \\ -6 & 9 & 39 \\ \hline 0 & 19 & 57 \end{array}\right\}$$

$$\sim \left(\begin{array}{cc|c} 1 & \frac{5}{3} & 3 \\ 0 & 1 & 3 \end{array}\right) \quad \frac{1}{19}R_2 \to R_2$$

From row 2, $y = 3$

Substituting into the first equation, $x + \frac{5}{3}(3) = 3$

$\therefore \quad x = -2$

So, the lines intersect at the point $(-2, 3)$.

Example 12 **Self Tutor**

Use row operations to solve $\begin{cases} x + 3y = 5 \\ 4x + 12y = k \end{cases}$ where k is a constant.

Give a geometric interpretation of the solutions.

In augmented matrix form, the system is:

$$\left(\begin{array}{cc|c} 1 & 3 & 5 \\ 4 & 12 & k \end{array}\right)$$

$$\sim \left(\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & 0 & k-20 \end{array}\right) \quad R_2 - 4R_1 \to R_2 \longleftarrow \left\{\begin{array}{ccc} 4 & 12 & k \\ -4 & -12 & -20 \\ \hline 0 & 0 & k-20 \end{array}\right\}$$

The second line reads $0x + 0y = k - 20$

Case 1: If $k \neq 20$ we have $0 =$ a non-zero number, which is impossible. There are no solutions, so the equations represent parallel lines.

Case 2: If $k = 20$ we have $0 = 0$, which is true for all x and y.

This means that the lines are coincident, and we can obtain the solutions from $x + 3y = 5$ alone.

Letting $y = t$, $x = 5 - 3t$ for all values of t.

$\therefore$ there are infinitely many solutions of the form $x = 5 - 3t$, $y = t$, $t \in \mathbb{R}$.

EXERCISE 13F.1

1 By inspection, classify the following pairs of equations as either intersecting, parallel, or coincident lines:

a $\begin{cases} x - 3y = 2 \\ 3x + y = 8 \end{cases}$ **b** $\begin{cases} x + y = 7 \\ 3x + 3y = 1 \end{cases}$ **c** $\begin{cases} 4x - y = 8 \\ y = 2 \end{cases}$

d $\begin{cases} x - 2y = 4 \\ 2x - 4y = 8 \end{cases}$ **e** $\begin{cases} 5x - 11y = 2 \\ 6x + y = 8 \end{cases}$ **f** $\begin{cases} 3x - 4y = 5 \\ -3x + 4y = 2 \end{cases}$

2 Reduce the system $\begin{cases} 4x + 5y = 15 \\ 3x - 2y = 17 \end{cases}$ using row reduction. Hence describe the relationship between the pair of lines.

3 Consider the equation pair $\begin{cases} x + 2y = 3 \\ 2x + 4y = 6 \end{cases}$.

a Explain why there are infinitely many solutions, giving geometric evidence.

b Explain why the second equation can be ignored when finding all solutions.

c Give all solutions in the form:

i $x = t,\ y = ...$ **ii** $y = s,\ x = ...$

4 **a** Use row reduction on the system $\begin{cases} 2x + 3y = 5 \\ 2x + 3y = 11 \end{cases}$ to reduce the system to $\left(\begin{array}{cc|c} 2 & 3 & 5 \\ 0 & 0 & 6 \end{array}\right)$.

b What does the second row indicate? Explain this result geometrically.

5 **a** Use row reduction on the system $\begin{cases} 2x + 3y = 5 \\ 4x + 6y = 10 \end{cases}$ to reduce the system to $\left(\begin{array}{cc|c} 2 & 3 & 5 \\ 0 & 0 & 0 \end{array}\right)$.

b Explain this result geometrically.

6 **a** By using augmented matrices, show that $\begin{cases} 3x - y = 2 \\ 6x - 2y = 4 \end{cases}$ has infinitely many solutions of the form $x = t,\ y = 3t - 2$.

b Discuss the solutions to $\begin{cases} 3x - y = 2 \\ 6x - 2y = k \end{cases}$ where k can take any real value.

7 Consider $\begin{cases} 3x - y = 8 \\ 6x - 2y = k \end{cases}$ where k can take any real value.

a Use row operations to reduce the system to the form: $\left(\begin{array}{cc|c} 3 & -1 & 8 \\ 0 & 0 & .... \end{array}\right)$.

b For what value(s) of k is there infinitely many solutions? Find the solutions in this case.

c For what value(s) of k does the system have no solutions? Explain this result geometrically.

8 Consider the system $\begin{cases} 4x + 8y = 1 \\ 2x - ay = 11 \end{cases}$ where a can take any real value.

a Use row operations to reduce the system to row echelon form.

b For what values of a does the system have a unique solution?

Show that the unique solution is $x = \dfrac{a + 88}{4a + 16},\ y = \dfrac{-21}{2a + 8}$ for these values of a.

c What is the solution in all other cases? Interpret the situation geometrically.

9 Consider the system $\begin{cases} mx + 2y = 6 \\ 2x + my = 6 \end{cases}$ where m can take any real value.

a Find the value(s) of m for which the system has a unique solution, and state the form of the unique solution. Interpret this solution geometrically.

b Discuss the solutions in the other *two* cases.

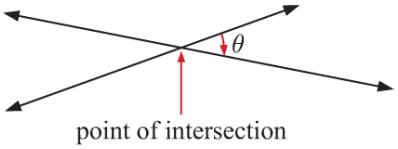
LINE CLASSIFICATION IN 3 DIMENSIONS

Lines in space are **coplanar** if they lie in the same plane. In this case they may be **intersecting**, **parallel**, or **coincident**.

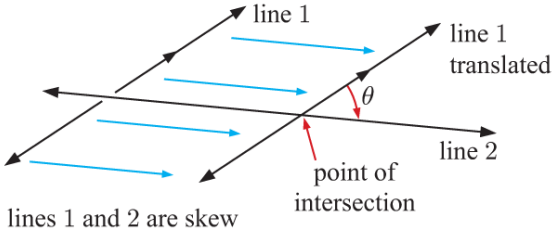
If the lines are not coplanar then they are **skew**.

We calculate the angle between two lines in space using the scalar product of the direction vectors of the lines.

- If the lines are **parallel**, the angle between them is $0°$.
- If the lines are **intersecting**, the angle between them is θ, as shown.



- If the lines are **skew**, we suppose one line is translated to intersect with the other. The angle between the original lines is defined as the angle between the intersecting lines, which is the angle θ.



Example 13 **Self Tutor**

Line 1 has equations $x = -1 + 2s,\ y = 1 - 2s,\ z = 1 + 4s,\ s \in \mathbb{R}$.

Line 2 has equations $x = 1 - t,\ y = t,\ z = 3 - 2t,\ t \in \mathbb{R}$.

Line 3 has equations $x = 1 + 2u,\ y = -1 - u,\ z = 4 + 3u,\ u \in \mathbb{R}$.

3D GRAPHING



a Show that line 1 and line 2 are parallel but not coincident.

b Show that line 2 and line 3 intersect and find the angle between them.

c Show that line 1 and line 3 are skew.

a Line 1 is $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + s\begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix}$ with direction vector $\begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix}$.

Line 2 is $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + t\begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}$ with direction vector $\begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}$.

Since $\begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix} = -2\begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}$, the lines are parallel.

If $\mathbf{a} = k\mathbf{b}$ for some scalar k, then $\mathbf{a} \parallel \mathbf{b}$.



When $s = 0$, the point on line 1 is $(-1,\ 1,\ 1)$.

For line 2, $y = t$.

$\therefore$ the unique point on line 2 with y-coordinate 1 is the point where $t = 1$. This point is $(0,\ 1,\ 1)$.

Since $(0,\ 1,\ 1) \neq (-1,\ 1,\ 1)$, the lines are not coincident.

b Equating x, y, and z values in lines 2 and 3 gives

$$1-t=1+2u, \quad t=-1-u \quad \text{and} \quad 3-2t=4+3u$$
$$\therefore\ t=-2u, \quad t=-1-u \quad \text{and} \quad 3u+2t=-1 \quad \ldots. (1)$$

Solving these we get $-2u=-1-u$

$$\therefore\ -u=-1$$
$$\therefore\ u=1 \text{ and so } t=-2$$

Checking in (1): $3u+2t=3(1)+2(-2)=3-4=-1$ ✓

Using $u=1$, lines 2 and 3 intersect at $(1+2(1), -1-(1), 4+3(1))$ which is $(3, -2, 7)$.

Direction vectors for lines 2 and 3 are $\mathbf{a}=\begin{pmatrix}-1\\1\\-2\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$ respectively.

If θ is the acute angle between $\mathbf{a}$ and $\mathbf{b}$, then

$$\cos\theta=\frac{|\mathbf{a}\bullet\mathbf{b}|}{|\mathbf{a}||\mathbf{b}|}=\frac{|-2-1-6|}{\sqrt{1+1+4}\sqrt{4+1+9}}=\frac{9}{\sqrt{84}}$$
$$\therefore\ \theta\approx 10.89^\circ$$

$\therefore$ the angle between lines 2 and 3 is about 10.9°.

c Equating x, y, and z values in lines 1 and 3 gives

$$-1+2s=1+2u \qquad 1-2s=-1-u \quad \text{and} \quad 1+4s=4+3u$$
$$\therefore\ 2s-2u=2 \qquad \therefore\ -2s+u=-2 \quad \text{and} \quad 4s-3u=3 \quad \ldots. (2)$$

Solving these we get $2s-2u=2$

$$-2s+u=-2$$

Adding, $-u=0$ and so $s=1$

Checking in (2): $4s-3u=4(1)-3(0)=4\neq 3$ ✗

There is no simultaneous solution to all 3 equations, so the lines do not intersect.

The direction vector for line 1 is $\begin{pmatrix}2\\-2\\4\end{pmatrix}$ and $\begin{pmatrix}2\\-2\\4\end{pmatrix}\neq k\begin{pmatrix}2\\-1\\3\end{pmatrix}$ for any $k\in\mathbb{R}$.

$\therefore$ lines 1 and 3 are not parallel.

Since the lines are not parallel and do not intersect, they are skew.

EXERCISE 13F.2

1 Classify the following line pairs as either parallel, intersecting, coincident, or skew. Where possible, state the point of intersection and find the measure of the acute angle between them.

a $x=1+2t, \quad y=2-t, \quad z=3+t$ and $x=-2+3s, \quad y=3-s, \quad z=1+2s$

b $x=-1+2\lambda, \quad y=2-12\lambda, \quad z=4+12\lambda$ and $x=4\mu-3, \quad y=3\mu+2, \quad z=-\mu-1$

c $x=6t, \quad y=3+8t, \quad z=-1+2t$ and $x=2+3s, \quad y=4s, \quad z=1+s$

d $x=2-y=z+2$ and $x=1+3s, \quad y=-2-2s, \quad z=2s+\frac{1}{2}$

e $x=1+\lambda, \quad y=2-\lambda, \quad z=3+2\lambda$ and $x=2+3\mu, \quad y=3-2\mu, \quad z=\mu-5$

f $x=1-2t, \quad y=8+t, \quad z=5$ and $x=2+4s, \quad y=-1-2s, \quad z=3$

g $x=1+2\lambda, \quad y=-\lambda, \quad z=1+3\lambda$ and $x=3-4\mu, \quad y=-1+2\mu, \quad z=4-6\mu$

2 Discuss the relationship between:

Line 1: $\mathbf{r}_1 = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$

Line 2: $x = 1 - 2\mu, \quad y = 4\mu, \quad z = -1 + 2\mu$

Line 3: $x = \dfrac{y-1}{2} = z - 1$

INVESTIGATION 1 MOTION OF PARTICLES IN SPACE

In this Investigation we consider the motion of two particles in space, and whether they will collide.

What to do:

1 Consider the lines L_1: $\begin{pmatrix} 5 \\ 4 \\ 4 \end{pmatrix} + t \begin{pmatrix} -2 \\ -3 \\ 4 \end{pmatrix}$, $t \in \mathbb{R}$ and L_2: $\begin{pmatrix} 1 \\ -4 \\ 14 \end{pmatrix} + u \begin{pmatrix} 0 \\ -2 \\ 2 \end{pmatrix}$, $u \in \mathbb{R}$.

Show that these lines intersect, and give the coordinates of the point of intersection.

2 Suppose we have two particles moving in space.

Particle A's position after t seconds is given by

$$x_A(t) = 5 - 2t, \quad y_A(t) = 4 - 3t, \quad z_A(t) = 4 + 4t.$$

Particle B's position after t seconds is given by

$$x_B(t) = 1, \quad y_B(t) = -4 - 2t, \quad z_B(t) = 14 + 2t.$$

All distance units are metres.

a Find the initial position of each particle.

b Find the velocity vector of each particle.

c Will the particles collide? Explain your answer.

3 Suppose D is the distance between particles A and B at time t.

a Write an equation connecting D and t.

b Hence find the minimum distance between the particles, and the time when this occurs.

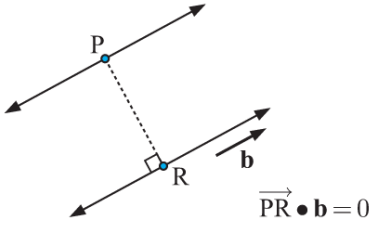
THE SHORTEST DISTANCE BETWEEN TWO LINES

We have seen that two lines in space may be *intersecting*, *parallel*, *coincident*, or *skew*.

If two lines in space intersect or are coincident, then clearly the shortest distance between them will be zero.

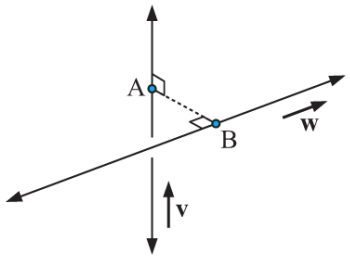
THE SHORTEST DISTANCE BETWEEN PARALLEL LINES

If the lines are **parallel**, we choose any point on one of the lines and find the shortest distance to the other line. We saw this in **Section D**.



THE SHORTEST DISTANCE BETWEEN SKEW LINES

If the lines are **skew**, we need to find two points A and B, one on each line, so that $\overrightarrow{AB}$ is perpendicular to *both* skew lines. So, if the direction vectors of the lines are $\mathbf{v}$ and $\mathbf{w}$ respectively, then $\overrightarrow{AB} \parallel \mathbf{v} \times \mathbf{w}$.



We write $\overrightarrow{AB}$ in terms of the parameters for the two lines, then use $\overrightarrow{AB} = k(\mathbf{v} \times \mathbf{w})$ to write a set of three equations in three unknowns. We solve these simultaneously to find the points A and B.

The shortest distance between the lines is $|\overrightarrow{AB}|$.

Example 14 **Self Tutor**

Find the shortest distance between the skew lines $x = t,\ y = 1 - t,\ z = 2 + t$ and $x = 3 - s,\ y = -1 + 2s,\ z = 4 - s$, where $s, t \in \mathbb{R}$.

The lines have direction vectors $\mathbf{v} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$

so $\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 1 \\ -1 & 2 & -1 \end{vmatrix} = \begin{vmatrix} -1 & 1 \\ 2 & -1 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix}\mathbf{k}$

$= -\mathbf{i} + \mathbf{k}$

Let A and B be points on the skew lines such that $|\overrightarrow{AB}|$ is the shortest distance between them.

$\therefore$ A is $(t,\ 1 - t,\ 2 + t)$ and B is $(3 - s,\ -1 + 2s,\ 4 - s)$ for some $s, t \in \mathbb{R}$.

Now $\overrightarrow{AB} \parallel \mathbf{v} \times \mathbf{w}$, so $\begin{pmatrix} 3 - s - t \\ -1 + 2s - 1 + t \\ 4 - s - 2 - t \end{pmatrix} = k \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$ for some $k \in \mathbb{R}$.

$\therefore\ 3 - s - t = -k$ .... (1)

$-2 + 2s + t = 0$ .... (2)

$2 - s - t = k$ .... (3)

Adding (1) and (3) gives $5 - 2s - 2t = 0$

$\therefore\ 2s + 2t = 5$

and rearranging (2) gives $-2s - t = -2$

Adding, $t = 3$

Using (2), $-2 + 2s + 3 = 0$ and so $s = -\frac{1}{2}$.

$\therefore\ \overrightarrow{AB} = \begin{pmatrix} 3 - \left(-\frac{1}{2}\right) - 3 \\ -2 + 2\left(-\frac{1}{2}\right) + 3 \\ 2 - \left(-\frac{1}{2}\right) - 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 0 \\ -\frac{1}{2} \end{pmatrix}$

and the shortest distance between the lines $= |\overrightarrow{AB}| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2}$

$= \frac{1}{\sqrt{2}}$ units.

EXERCISE 13F.3

1 Find the shortest distance between the skew lines:

a $x = 1 + 2t,\ y = -t,\ z = 2 + 3t,\ t \in \mathbb{R}$ and $x = y = z$

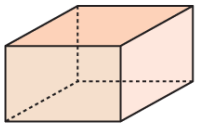
b $x = 1 - t,\ y = 1 + t,\ z = 3 - t$ and $x = 2 + s,\ y = 1 - 2s,\ z = s,$ where $s, t \in \mathbb{R}$.

2 Find the shortest distance between each pair of lines in **Exercise 13F.2** question **1**.

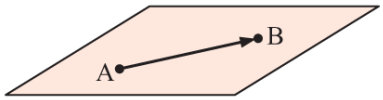
G PLANES

A **plane** is a flat surface that extends forever, and has zero thickness.

It fits into a scheme which starts with a point and which builds to three dimensional space.

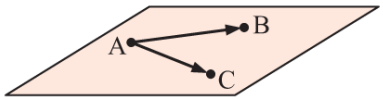
point	**line**	**plane**	**space**
0 dimensions	1 dimension	2 dimensions	3 dimensions
•			

Suppose two points A and B lie on a plane. We say that $\overrightarrow{AB}$ is *parallel* to the plane.



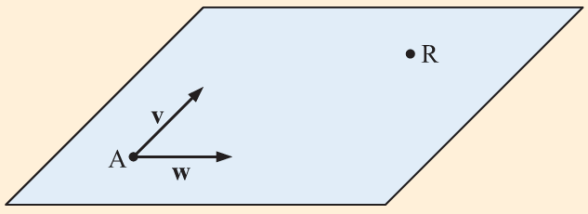
A vector is **parallel** to a plane if, by travelling along the vector, we remain on the plane.

In the diagram alongside, we can see two vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ which are both parallel to the plane, but *not* parallel to each other.



INVESTIGATION 2 LINEAR COMBINATIONS

Suppose we are at a fixed point A with position vector **a** on a plane. Also suppose two non-parallel vectors **v** and **w** are both parallel to the plane.



In this Investigation we consider travelling from A to another point R using a **linear combination** of **v** and **w**, which means adding scalar multiples of **v** and **w** to the position vector of A. If the position vector of R is **r**, then $\mathbf{r} = \mathbf{a} + s\mathbf{v} + t\mathbf{w}$ for some $s, t \in \mathbb{R}$.

What to do:

1 Explain why:

a the vector $\mathbf{a} + s\mathbf{v}$ lies on the plane for any $s \in \mathbb{R}$

b the vector $\mathbf{a} + s\mathbf{v} + t\mathbf{w}$ lies on the plane for any $s, t \in \mathbb{R}$

c it is important that $\mathbf{v}$ and $\mathbf{w}$ are non-parallel.

2 Consider two lines on the plane:

Line 1 has vector equation $\mathbf{a} + s\mathbf{v}$, $s \in \mathbb{R}$.

Line 2 has vector equation $\mathbf{r} - t\mathbf{w}$, $t \in \mathbb{R}$.

a Draw a vector diagram to illustrate this situation.

b Write a vector equation for the intersection of the two lines.

c Explain how the lines and the vector equation relate to our problem of moving from A to R using a linear combination of $\mathbf{v}$ and $\mathbf{w}$.

d When we solve the vector equation to find s and t, explain why there will be a unique solution.

3 **a** Can *any* point on the plane be reached from A using a linear combination of $\mathbf{v}$ and $\mathbf{w}$?

b Can any point *not* on the plane be reached from A using a linear combination of $\mathbf{v}$ and $\mathbf{w}$?

THE VECTOR EQUATION OF A PLANE

From the **Investigation**, you should have found that any point on a plane can be reached from any other using a unique linear combination of any two non-parallel vectors which are parallel to the plane.

$\mathbf{r} = \mathbf{a} + s\mathbf{v} + t\mathbf{w}$ is the **vector equation of a plane**

where $\mathbf{r}$ is the position vector of any point on the plane,

$\mathbf{a}$ is the position vector of a known point on the plane

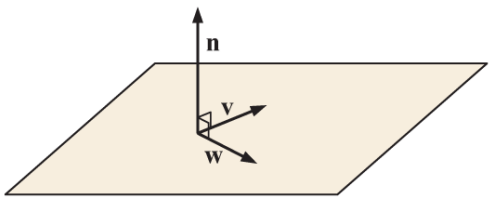
$\mathbf{v}$ and $\mathbf{w}$ are any two non-parallel vectors that are parallel to the plane

$s, t \in \mathbb{R}$ are two independent parameters.

THE NORMAL OF A PLANE

A vector is **normal** to a plane if it is perpendicular to all vectors which are parallel to the plane.

For example, if $\mathbf{v}$ and $\mathbf{w}$ are any two non-parallel vectors on a plane, then $\mathbf{n} = \mathbf{v} \times \mathbf{w}$ is normal to the plane.

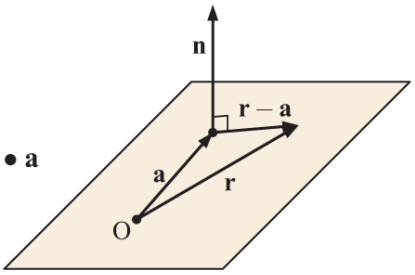


We can also write the vector equation of a plane using a normal vector.

Suppose $\mathbf{r}$ is the position vector of any point on a plane
and $\mathbf{a}$ is the position vector of a known point on the plane.

Since $\mathbf{n}$ is perpendicular to any vector in the plane, $\mathbf{n} \bullet (\mathbf{r} - \mathbf{a}) = 0$

$\therefore \quad \mathbf{n} \bullet \mathbf{r} = \mathbf{n} \bullet \mathbf{a}$



The normal equation of a plane is $\mathbf{r} \bullet \mathbf{n} = \mathbf{a} \bullet \mathbf{n}$.

THE CARTESIAN EQUATION OF A PLANE

We can use this form of the vector equation of a plane to derive a Cartesian form:

If a plane has normal vector $\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ and passes through (X, Y, Z) then it has equation $ax + by + cz = aX + bY + cZ = d$, where d is a constant.

This is the **Cartesian equation of the plane**.

Example 15 **Self Tutor**

Find the equation of the plane with normal vector $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and which passes through the point $(-1, 2, 4)$.

$\mathbf{n} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, and the point $(-1, 2, 4)$ lies on the plane.

$\therefore$ the equation is $x + 2y + 3z = (-1) + 2(2) + 3(4)$

which is $x + 2y + 3z = 15$

EXERCISE 13G

1 Find the Cartesian equation of the plane:

a with normal vector $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and which passes through $(-1, 2, 4)$

b perpendicular to the line connecting A$(2, 3, 1)$ and B$(5, 7, 2)$, and which passes through A.

2 State the normal vector to the plane with equation:

a $2x + 3y - z = 8$ **b** $3x - y = 11$ **c** $z = 2$ **d** $x = 0$

3 Find the Cartesian equation of the:

a XZ-plane

b plane perpendicular to the Z-axis, which passes through $(2, -1, 4)$.

Example 16 **Self Tutor**

Find the equation of the plane through A$(-1, 2, 0)$, B$(3, 1, 1)$, and C$(1, 0, 3)$:

a in vector form **b** in Cartesian form.

a $\overrightarrow{AB} = \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$ and $\overrightarrow{CB} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ are two non-parallel vectors in the plane.

Using C as the known (fixed) point on the plane,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + s\begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix} + t\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, \quad s, t \in \mathbb{R}.$$

b The normal vector $\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$

$$\therefore \mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -1 & 1 \\ 2 & -2 & 3 \end{vmatrix} = \begin{vmatrix} -1 & 1 \\ -2 & 3 \end{vmatrix}\mathbf{i} - \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix}\mathbf{j} + \begin{vmatrix} 4 & -1 \\ 2 & -2 \end{vmatrix}\mathbf{k}$$

$$= -1\mathbf{i} - 10\mathbf{j} - 6\mathbf{k}$$

$$= -(\mathbf{i} + 10\mathbf{j} + 6\mathbf{k})$$

$\therefore$ the vector $\begin{pmatrix} 1 \\ 10 \\ 6 \end{pmatrix}$ is also normal to the plane.

Thus the plane has equation

$x + 10y + 6z = 1(-1) + 10(2) + 6(0)$ {using point A}

$\therefore \; x + 10y + 6z = 19$



Check that the other two points also satisfy the equation.

4 Find the equation of the plane through A, B, and C, giving your answer in:

i vector form **ii** Cartesian form.

a A$(0, 2, 6)$, B$(1, 3, 2)$, and C$(-1, 2, 4)$ **b** A$(3, 1, 2)$, B$(0, 4, 0)$, and C$(0, 0, 1)$

c A$(2, 0, 3)$, B$(0, -1, 2)$, and C$(4, -3, 0)$.

5 A plane has equation $\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} + s\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} + t\begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix}$, $s, t \in \mathbb{R}$. Write the equation of the plane in Cartesian form.

6 Find the Cartesian equation of the plane containing A$(3, 2, 1)$ and the line $x = 1 + t$, $y = 2 - t$, $z = 3 + 2t$, $t \in \mathbb{R}$.

7 Find the Cartesian equation of the plane through A$(1, 2, 3)$ and B$(0, -1, 2)$ which is parallel to:

a the X-axis **b** the Y-axis **c** the Z-axis.

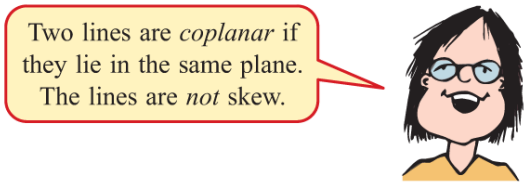
8 Find the vector equation of the line:

a which is normal to the plane $x - 3y + 4z = 8$ and passes through $(1, -2, 0)$

b which is normal to the plane $x - y - 2z = 11$ and passes through $(3, 4, -1)$.

9 Does the line passing through $(3, 4, -1)$ and which is normal to $x + 4y - z = -2$, intersect any of the coordinate axes?

10 Show that the lines $x - 1 = \dfrac{y-2}{2} = z + 3$ and $x + 1 = y - 3 = 2z + 5$ are coplanar, and find the Cartesian equation of the plane which contains them.



11 The equations of two lines are:

$$L_1: \quad \mathbf{r} = \begin{pmatrix} -4 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}, \ \lambda \in \mathbb{R} \quad \text{and} \quad L_2: \quad x = \frac{y-5}{2} = \frac{-z-1}{2}.$$

a Find the point of intersection of L_1 and L_2.

b Find the equation of the plane that contains L_1 and L_2.

12 A$(1, 2, k)$ lies on the plane $x + 2y - 2z = 8$. (AB) is normal to the plane, and B is 6 units from it.

a Find the value of k.

b Find the coordinates of B.

Example 17 **Self Tutor**

Find the parametric equations of the line through A$(-1, 2, 3)$ and B$(2, 0, -3)$.

Hence find where this line meets the plane with equation $x - 2y + 3z = 26$.

$\overrightarrow{\text{AB}} = \begin{pmatrix} 3 \\ -2 \\ -6 \end{pmatrix}$ so (AB) has parametric equations

$$x = -1 + 3t, \quad y = 2 - 2t, \quad z = 3 - 6t, \quad t \in \mathbb{R} \quad \ldots. \ (*)$$

This line meets the plane $x - 2y + 3z = 26$ where

$$-1 + 3t - 2(2 - 2t) + 3(3 - 6t) = 26$$
$$\therefore \ 4 - 11t = 26$$
$$\therefore \ -11t = 22$$
$$\therefore \ t = -2$$

Substituting $t = -2$ into $(*)$, the line meets the plane at $(-7, 6, 15)$.

13 Find the parametric equations of the line through A$(2, -1, 3)$ and B$(1, 2, 0)$.
Hence find where this line meets the plane with equation $x + 2y - z = 21$.

14 **a** Find the parametric equations of the line through P$(1, -2, 4)$ and Q$(2, 0, -1)$.

b Hence find where this line meets:

i the YZ-plane

ii the plane with equation $y + z = 2$

iii the line with Cartesian equation $\dfrac{x-3}{2} = \dfrac{y+2}{3} = \dfrac{z-30}{-1}$.

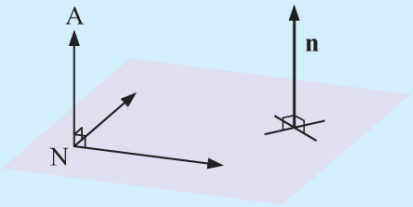
15 A plane P has equation $2x - 5y + z = 10$. A line L is normal to P, and passes through $(1, 5, 3)$.

a Find the parametric equations of L.

b Find the point at which L meets P.

Example 18 **Self Tutor**

Suppose N is the foot of the normal from $A(2, -1, 3)$ to the plane $x - y + 2z = 27$. Find the coordinates of N, and hence find the shortest distance from A to the plane.



$x - y + 2z = 27$ has normal vector $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$

$\therefore$ the parametric equations of (AN) are:

$x = 2 + t, \ y = -1 - t, \ z = 3 + 2t, \ t \in \mathbb{R}$.

This line meets the plane $x - y + 2z = 27$ where

$2 + t - (-1 - t) + 2(3 + 2t) = 27$

$\therefore \ 2 + t + 1 + t + 6 + 4t = 27$

$\therefore \ 6t + 9 = 27$

$\therefore \ 6t = 18$

$\therefore \ t = 3$

Since N lies on the plane, its coordinates must satisfy the equation of the plane.



$\therefore$ N has coordinates $(5, -4, 9)$.

The shortest distance, $|\overrightarrow{AN}| = \sqrt{(5-2)^2 + (-4-(-1))^2 + (9-3)^2}$

$= \sqrt{54}$ units.

16 Find the foot of the normal from A to the given plane, and hence find the shortest distance from A to the plane:

a $A(1, 0, 2), \quad 2x + y - 2z + 11 = 0$

b $A(2, -1, 3), \quad x - y + 3z = -10$

c $A(1, -4, -3), \quad 4x - y - 2z = 8$

17 Find the coordinates of the mirror image of $A(3, 1, 2)$ when reflected in the plane $x + 2y + z = 1$.

18 A plane P has equation $\mathbf{r} = 2\mathbf{i} - 4\mathbf{k} + s(2\mathbf{i} - \mathbf{j} + \mathbf{k}) + t(\mathbf{i} + \mathbf{j} - 2\mathbf{k})$ where $s, t \in \mathbb{R}$.

a Write the equation of P in Cartesian form.

b Find the coordinates of the foot of the normal from $A(5, 8, 5)$ to P.

19 In each of the following cases, find the foot of the normal from A to the given plane, and hence find the shortest distance from A to the plane:

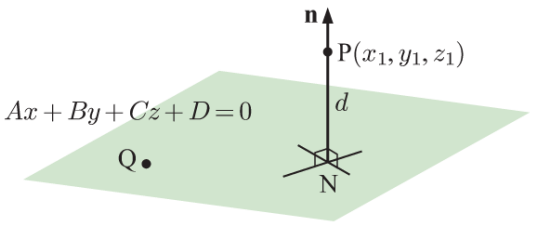
a $A(3, 2, 1), \quad \mathbf{r} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k} + \lambda(2\mathbf{i} + \mathbf{j} + \mathbf{k}) + \mu(4\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}), \ \lambda, \mu \in \mathbb{R}$

b $A(1, 0, -2), \quad \mathbf{r} = \mathbf{i} - \mathbf{j} + \mathbf{k} + \lambda(3\mathbf{i} - \mathbf{j} + 2\mathbf{k}) + \mu(-\mathbf{i} + \mathbf{j} - \mathbf{k}), \ \lambda, \mu \in \mathbb{R}$

20 Suppose $\mathbf{v}$ and $\mathbf{w}$ are non-parallel vectors which lie in the same plane. The normal vector to the plane is $\mathbf{n} = \mathbf{v} \times \mathbf{w}$.

Prove that $\mathbf{n} \perp (s\mathbf{v} + t\mathbf{w})$ for all $s, t \in \mathbb{R}$ except when $s = t = 0$.

21 Suppose Q is any point in the plane $Ax + By + Cz + D = 0$. Let d be the distance from $P(x_1, y_1, z_1)$ to the given plane.



a Explain why $d = \dfrac{|\overrightarrow{QP} \bullet \mathbf{n}|}{|\mathbf{n}|}$.

b Hence show that $d = \dfrac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$.

c Check your answers to question **16** using the formula in **b**.

d Find the distance from:

i $(0, 0, 0)$ to $x + 2y - z = 10$ **ii** $(1, -3, 2)$ to $x + y - z = 2$.

22 Find the distance between the parallel planes:

a $x + y + 2z = 4$ and $2x + 2y + 4z + 11 = 0$

b $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$.

23 Show that the line $x = 2 + t$, $y = -1 + 2t$, $z = -3t$ is parallel to the plane $11x - 4y + z = 0$, and find its distance from the plane.

24 Find the Cartesian equations of the two planes which are parallel to $2x - y + 2z = 5$ and 2 units from it.

In questions **22**, **23**, **24**, use the formula $d = \dfrac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$.



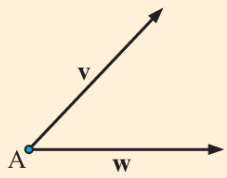
INVESTIGATION 3 REGIONS OF A PLANE

We have previously seen that the position of any point on a plane can be described by $\mathbf{r} = \mathbf{a} + s\mathbf{v} + t\mathbf{w}$ where $\mathbf{a}$ is the position vector of a fixed point on the plane, and $s\mathbf{v} + t\mathbf{w}$ is a linear combination of any two non-parallel vectors which are parallel to the plane.

We will now investigate *regions* of a plane by restricting the possible values of the parameters s and t.

What to do:

1



a Copy the diagram alongside. Shade the region which includes all points $\mathbf{r} = \mathbf{a} + s\mathbf{v} + t\mathbf{w}$ where $\mathbf{a}$ is the position vector of point A, $0 \leqslant s \leqslant 1$, and $0 \leqslant t \leqslant 1$.

b Describe the shape of the region.

c What would the shape of the region be for $0 \leqslant s \leqslant 2$ and $0 \leqslant t \leqslant 1$?

2 Suppose $\mathbf{v}$ and $\mathbf{w}$ are perpendicular and that $|\mathbf{v}| = |\mathbf{w}|$. Using a sketch to help you if necessary, describe the shape of the region formed on the plane $\mathbf{r} = \mathbf{a} + s\mathbf{v} + t\mathbf{w}$ for:

a $0 \leqslant s \leqslant 1$ and $0 \leqslant t \leqslant 1$ **b** $0 \leqslant s \leqslant 2$ and $0 \leqslant t \leqslant 1$

3 Consider the range of values $s^2 + t^2 \leqslant 1$. Describe the shape of the region formed on the plane $\mathbf{r} = \mathbf{a} + s\mathbf{v} + t\mathbf{w}$ if $\mathbf{v}$ and $\mathbf{w}$ are:

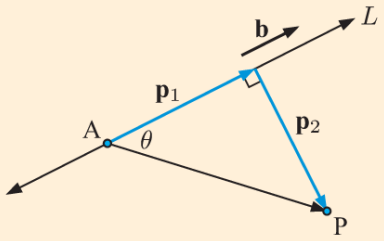
a perpendicular **b** neither parallel nor perpendicular.

INVESTIGATION 4 SHORTEST DISTANCE FORMULAE

The process of finding the shortest distance from a point to a line or plane, or between two skew lines, can often be long or difficult. In this Investigation, we will consider an alternate approach to these types of problems.

What to do:

1 Suppose the line L: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, $\lambda \in \mathbb{R}$ passes through a point A. Let P be a point that does *not* lie on L.



$\overrightarrow{AP}$ can be *decomposed* into a vector $\mathbf{p}_1$ which is parallel to L, and a vector $\mathbf{p}_2$ which is perpendicular to L as shown.

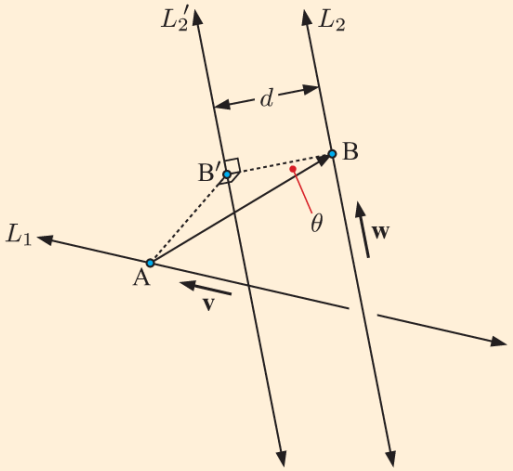
a Explain why the shortest distance from P to L is given by $|\,\mathbf{p}_2\,|$.

b Hence show that the shortest distance from P to L is given by $\dfrac{|\,\overrightarrow{AP} \times \mathbf{b}\,|}{|\,\mathbf{b}\,|}$.

c Explain why the expression is only useful in 3-dimensional space.

2 Consider the skew lines L_1: $\mathbf{r}_1 = \mathbf{a} + \lambda\mathbf{v}$ and L_2: $\mathbf{r}_2 = \mathbf{b} + \mu\mathbf{w}$, $\lambda, \mu \in \mathbb{R}$. Let A and B be points on L_1 and L_2 respectively.

Suppose L_2 is translated onto a line L_2' which intersects with L_1, so L_2' and L_1 lie on the same plane and B′ is the foot of the perpendicular from B to this plane.



a Explain why $\overrightarrow{B'B}$ is a scalar multiple of $\mathbf{v} \times \mathbf{w}$.

b Hence show that the shortest distance between L_1 and L_2 is given by

$$d = \frac{|\,\overrightarrow{AB} \bullet (\mathbf{v} \times \mathbf{w})\,|}{|\,\mathbf{v} \times \mathbf{w}\,|}.$$

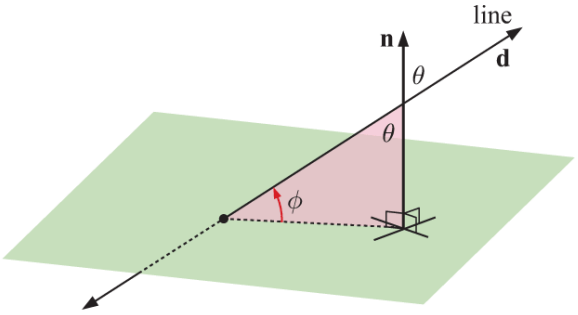
c Explain why the above expression cannot be used to find the distance between two non-coincident parallel lines.

d Check your answers to **Exercise 13F.3** using the expression you derived in **b**.

H ANGLES IN SPACE

THE ANGLE BETWEEN A LINE AND A PLANE

Suppose a line has direction vector $\mathbf{d}$ and a plane has normal vector $\mathbf{n}$. We allow $\mathbf{n}$ to intersect the line, making an angle of θ with it. The angle between the line and the plane is the acute angle ϕ.



Now $\sin\phi = \cos\theta = \dfrac{|\mathbf{n} \bullet \mathbf{d}|}{|\mathbf{n}||\mathbf{d}|}$

$$\therefore \quad \phi = \sin^{-1}\left(\frac{|\mathbf{n} \bullet \mathbf{d}|}{|\mathbf{n}||\mathbf{d}|}\right)$$

Example 19 Self Tutor

Find the angle between the plane $x + 2y - z = 8$ and the line with equations $x = t,\ y = 1 - t,\ z = 3 + 2t,\ t \in \mathbb{R}$.

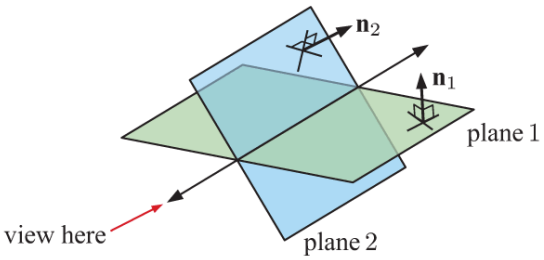
$\mathbf{n} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{d} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$

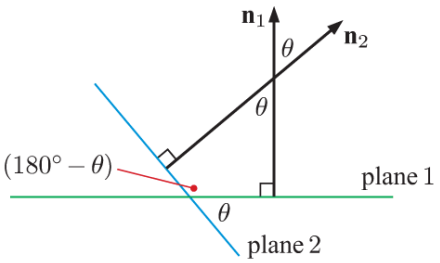
The angle between the plane and the line is $\phi = \sin^{-1}\left(\dfrac{|1 - 2 - 2|}{\sqrt{1+4+1}\sqrt{1+1+4}}\right)$

$= \sin^{-1}\left(\dfrac{3}{\sqrt{6}\sqrt{6}}\right)$

$= \sin^{-1}\left(\frac{1}{2}\right) = 30°$

THE ANGLE BETWEEN TWO PLANES



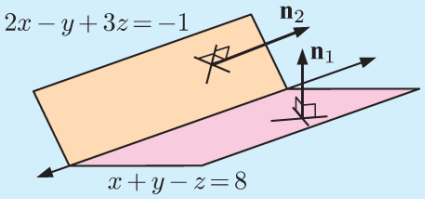


The cosine of the acute angle between the two normals is $\cos\theta = \dfrac{|\mathbf{n}_1 \bullet \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}$.

If two planes have normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, and θ is the acute angle between them, then $\theta = \cos^{-1}\left(\dfrac{|\mathbf{n}_1 \bullet \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}\right)$.

Example 20 **Self Tutor**

Find the angle between the planes with equations $x + y - z = 8$ and $2x - y + 3z = -1$.



$x + y - z = 8$ has normal vector $\mathbf{n}_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$.

$2x - y + 3z = -1$ has normal vector $\mathbf{n}_2 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$.

The acute angle between the planes is $\theta = \cos^{-1}\left(\dfrac{|\mathbf{n}_1 \bullet \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|}\right)$

$= \cos^{-1}\left(\dfrac{|2 - 1 - 3|}{\sqrt{1+1+1}\sqrt{4+1+9}}\right)$

$= \cos^{-1}\left(\dfrac{2}{\sqrt{42}}\right)$

$\approx 72.0°$

EXERCISE 13H

1 Find the angle between:

a the plane $x - y + z = 5$ and the line $\dfrac{x-1}{4} = \dfrac{y+1}{3} = z + 2$

b the plane $2x - y + z = 8$ and the line $x = t + 1$, $y = -1 + 3t$, $z = t$, $t \in \mathbb{R}$

c the plane $3x + 4y - z = -4$ and the line $x - 4 = 3 - y = 2(z + 1)$

d the plane $\mathbf{r}_p = 2\mathbf{i} - \mathbf{j} + \mathbf{k} + \lambda(3\mathbf{i} - 4\mathbf{j} - \mathbf{k}) + \mu(\mathbf{i} + \mathbf{j} - 2\mathbf{k})$, $\lambda, \mu \in \mathbb{R}$ and the line $\mathbf{r}_l = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k} + t(\mathbf{i} - \mathbf{j} + \mathbf{k})$, $t \in \mathbb{R}$.

2 **a** Show that the line $x = 1 + 3t$, $y = 4 - t$, $z = 2 + 4t$, $t \in \mathbb{R}$ is parallel to the plane $2x + 2y - z = 5$.

b Find the distance between the line and the plane.

3 Find the angle between the plane $3x - y + 4z = -7$ and the:

a X-axis **b** Y-axis **c** Z-axis.

4 The line L passes through A$(-3,\ 2,\ 5)$ and B$(0,\ 2,\ -1)$, and the plane π_1 has equation $4x - 2y + 5z = 16$.

a Find the angle between L and π_1.

b Find the coordinates of C, the point at which π_1 cuts the Y-axis.

c The plane π_2 is perpendicular to (AC) and passes through B.

i Find the equation of π_2. **ii** Find the angle between L and π_2.

5 Find the acute angle between the planes with equations:

a $2x - y + z = 3$, $x + 3y + 2z = 8$

b $x - y + 3z = 2$, $3x + y - z = -5$

c $3x - y + z = -11$, $2x + 4y - z = 2$

d $\mathbf{r}_1 = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k} - \lambda(\mathbf{i} - \mathbf{j} + \mathbf{k}) + \mu(2\mathbf{i} - 4\mathbf{j} + 3\mathbf{k})$, $\lambda, \mu \in \mathbb{R}$
$\mathbf{r}_2 = \mathbf{i} + \mathbf{j} - \mathbf{k} - \lambda(2\mathbf{i} + \mathbf{j} + \mathbf{k}) + \mu(\mathbf{i} + \mathbf{j} + \mathbf{k})$, $\lambda, \mu \in \mathbb{R}$

e $3x - 4y + z = -2$ and $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$, $\lambda, \mu \in \mathbb{R}$

6 The planes $x + 5y - 3z = 8$ and $2x + 2y + kz = 9$ are perpendicular. Find the value of k.

7 The plane π_1 contains lines L_1 and L_2 which intersect.

L_1 has equation $\mathbf{r} = \begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$, $t \in \mathbb{R}$, and

L_2 has equation $\mathbf{r} = \begin{pmatrix} 8 \\ -3 \\ k \end{pmatrix} + s \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix}$, $s \in \mathbb{R}$.

a Find k and the point of intersection.

b Find, in Cartesian form, the equation of π_1.

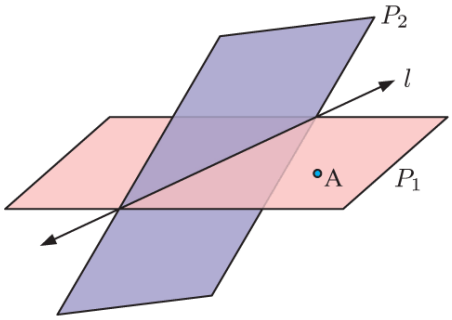
c The plane π_2 has equation $2y + 11z = -7$. Show that π_1 and π_2 are perpendicular.

d Find the angle between:

i L_1 and L_2 **ii** L_1 and π_2 **iii** L_2 and π_2.

e Find the sum of the angles in **d**. Draw a diagram to help explain your result.

8 P_1 has equation $2x - y + 3z = 7$, and P_2 has equation $x - 4y + z = 23$.



a Find the angle between P_1 and P_2.

b A$(-2, 4, k)$ lies on P_1. Find the value of k.

c A line through A, normal to P_1, cuts P_2 at B. Find the coordinates of B.

d l is the line of intersection of P_1 and P_2.

i Show that $(-4, -6, 3)$ lies on l.

ii Explain why l has direction $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \times \begin{pmatrix} 1 \\ -4 \\ 1 \end{pmatrix}$.

e The plane P_3 contains l and the midpoint of [AB]. Find the equation of P_3.

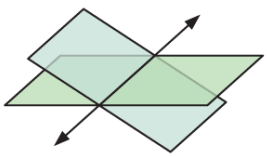
f Find the angle between: **i** P_1 and P_3 **ii** P_3 and P_2.

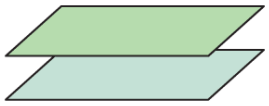
g Find the sum of the angles in **f**. Comment on your answer.

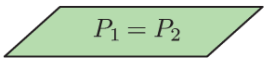
I INTERSECTING PLANES

- **Two planes** in space could have any of the following three arrangements:

(1) intersecting **(2)** parallel **(3)** coincident

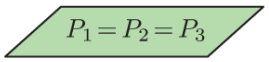




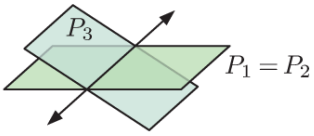


- **Three planes** in space could have any of the following eight arrangements:

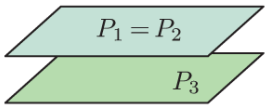
(1) all coincident



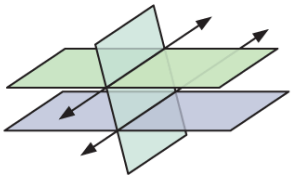
(2) two coincident and one intersecting



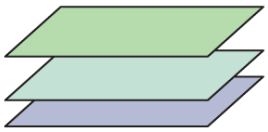
(3) two coincident and one parallel



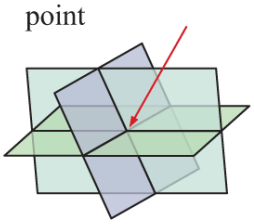
(4) two parallel and one intersecting



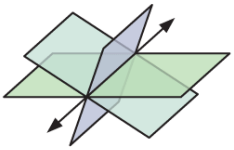
(5) all three parallel



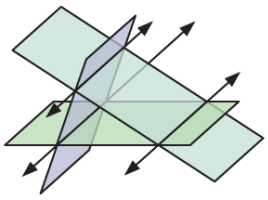
(6) all meet at the one point



(7) all meet in a common line



(8) the line of intersection of any two is parallel to the third plane.



Click on the icon to explore these cases in three dimensions.



To determine the relationship between planes, we need to solve a system of linear equations. We do this using row operations.

Example 21 **Self Tutor**

Solve the following systems of linear equations. In each case give a geometric interpretation of the system.

a $\begin{cases} x + 3y - z = 15 \\ 2x + y + z = 7 \\ x - y - 2z = 0 \end{cases}$ **b** $\begin{cases} 2x - y + z = 5 \\ x + y - z = 2 \\ 3x - 3y + 3z = 8 \end{cases}$

a In augmented matrix form, the system is:

$$\left(\begin{array}{ccc|c} 1 & 3 & -1 & 15 \\ 2 & 1 & 1 & 7 \\ 1 & -1 & -2 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3 & -1 & 15 \\ 0 & -5 & 3 & -23 \\ 0 & -4 & -1 & -15 \end{array}\right) \quad \begin{array}{l} R_2 - 2R_1 \to R_2 \\ R_3 - R_1 \to R_3 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3 & -1 & 15 \\ 0 & 1 & -\frac{3}{5} & \frac{23}{5} \\ 0 & 0 & -17 & 17 \end{array}\right) \quad \begin{array}{l} -\frac{1}{5}R_2 \to R_2 \\ 5R_3 - 4R_2 \to R_3 \end{array}$$

$$\left\{\begin{array}{cccc} 2 & 1 & 1 & 7 \\ -2 & -6 & 2 & -30 \\ \hline 0 & -5 & 3 & -23 \end{array}\right\}$$

$$\left\{\begin{array}{cccc} 1 & -1 & -2 & 0 \\ -1 & -3 & 1 & -15 \\ \hline 0 & -4 & -1 & -15 \end{array}\right\}$$

$$\left\{\begin{array}{cccc} 0 & -20 & -5 & -75 \\ 0 & 20 & -12 & 92 \\ \hline 0 & 0 & -17 & 17 \end{array}\right\}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 3 & -1 & 15 \\ 0 & 1 & -\frac{3}{5} & \frac{23}{5} \\ 0 & 0 & 1 & -1 \end{array}\right) \quad -\tfrac{1}{17}R_3 \to R_3$$

The last row gives $z = -1$

Using the final row 2, $y - \frac{3}{5}(-1) = \frac{23}{5}$

$\therefore \ y = 4$

Using the final row 1, $x + 3(4) - (-1) = 15$

$\therefore \ x = 2$

Thus we have a unique solution $x = 2, \ y = 4, \ z = -1$.

The system represents three planes that meet at the single point $(2, 4, -1)$. {case **(6)**}

b In augmented matrix form, the system is:

$$\left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & -1 & 1 & 5 \\ 3 & -3 & 3 & 8 \end{array}\right) \quad R_1 \leftrightarrow R_2 \qquad \left\{\begin{array}{cccc} 2 & -1 & 1 & 5 \\ -2 & -2 & 2 & -4 \\ \hline 0 & -3 & 3 & 1 \end{array}\right\}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -3 & 3 & 1 \\ 0 & -6 & 6 & 2 \end{array}\right) \quad \begin{array}{l} R_2 - 2R_1 \to R_2 \\ R_3 - 3R_1 \to R_3 \end{array} \qquad \left\{\begin{array}{cccc} 3 & -3 & 3 & 8 \\ -3 & -3 & 3 & -6 \\ \hline 0 & -6 & 6 & 2 \end{array}\right\}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -1 & -\frac{1}{3} \\ 0 & 0 & 0 & 0 \end{array}\right) \quad \begin{array}{l} -\frac{1}{3}R_2 \to R_2 \\ R_3 - 2R_2 \to R_3 \end{array} \qquad \left\{\begin{array}{cccc} 0 & -6 & 6 & 2 \\ 0 & 6 & -6 & -2 \\ \hline 0 & 0 & 0 & 0 \end{array}\right\}$$

The row of zeros indicates infinitely many solutions.

If we let $z = t$ in row 2, then $y - t = -\frac{1}{3}$

$\therefore \ y = -\frac{1}{3} + t$

Thus in row 1, $x + (-\frac{1}{3} + t) - t = 2$

$\therefore \ x - \frac{1}{3} = 2$ and so $x = \frac{7}{3}$

$\therefore$ the solutions have the form: $x = \frac{7}{3}, \ y = -\frac{1}{3} + t, \ z = t$, where $t \in \mathbb{R}$.

The system represents three planes that intersect in a line. {case **(7)**}

EXERCISE 13I

1 **a** How many geometric interpretations are possible when solving $\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \end{cases}$ simultaneously?

b Under what conditions will the planes in **a** be:

i parallel **ii** coincident?

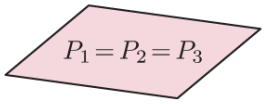
c Solve the following using elementary row operations, and interpret each system of equations geometrically:

i $\begin{cases} x - 3y + 2z = 8 \\ 3x - 9y + 2z = 4 \end{cases}$ **ii** $\begin{cases} 2x + y + z = 5 \\ x - y + z = 3 \end{cases}$ **iii** $\begin{cases} x + 2y - 3z = 6 \\ 3x + 6y - 9z = 18 \end{cases}$

2 Discuss the possible solutions of the following systems where k is a real number, and interpret the solutions geometrically.

a $\begin{cases} x+2y-z=6 \\ 2x+4y+kz=12 \end{cases}$ **b** $\begin{cases} x-y+3z=8 \\ 2x-2y+6z=k \end{cases}$

3 For each of the eight possible geometric scenarios of three planes in space, comment on the solutions to the corresponding system of linear equations.

For example, 

has infinitely many solutions where x, y, and z are expressed in terms of two parameters s and t.

4 Solve the following systems using row operations. State the geometric meaning of each solution.

a $\begin{cases} x+y-z=-5 \\ x-y+2z=11 \\ 4x+y-5z=-18 \end{cases}$ **b** $\begin{cases} x-y+2z=1 \\ 2x+y-z=8 \\ 5x-2y+5z=11 \end{cases}$ **c** $\begin{cases} x+2y-z=8 \\ 2x-y-z=5 \\ 3x-4y-z=2 \end{cases}$

d $\begin{cases} x-y+z=8 \\ 2x-2y+2z=11 \\ x+3y-z=-2 \end{cases}$ **e** $\begin{cases} x+y-2z=1 \\ x-y+z=4 \\ 3x+3y-6z=3 \end{cases}$ **f** $\begin{cases} x-y-z=5 \\ x+y+z=1 \\ 5x-y+2z=17 \end{cases}$

5 **a** Solve the system of equations $\begin{cases} 2x-y+4z=5 \\ x+y-2z=-3 \\ -x-4y+10z=6 \end{cases}$.

Give a geometric interpretation of your solution.

b Find the equation of l, the line of intersection of the planes $2x-y+4z=5$ and $x+y-2z=-3$.

c Verify that l is parallel to the plane $-x-4y+10z=6$.

Example 22 **Self Tutor**

Consider the system $\begin{cases} x+y+3z=2 \\ 3x-y+2z=5 \\ 5x+y+8z=a \end{cases}$ where a could take any real value.

a Use row operations to reduce the system to row echelon form.

b For what value(s) of a does the system have no solutions? Interpret the result geometrically.

c For what value(s) of a does the system have infinitely many solutions? Find the solutions in this case, and interpret them geometrically.

a In augmented matrix form, the system is:

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 3 & -1 & 2 & 5 \\ 5 & 1 & 8 & a \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & -4 & -7 & -1 \\ 0 & -4 & -7 & a-10 \end{array}\right) \begin{array}{l} R_2 - 3R_1 \to R_2 \\ R_3 - 5R_1 \to R_3 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & \frac{7}{4} & \frac{1}{4} \\ 0 & 0 & 0 & a-9 \end{array}\right) \begin{array}{l} \\ -\frac{1}{4}R_2 \to R_2 \\ R_3 - R_2 \to R_3 \end{array}$$

b If $a \neq 9$, we have an inconsistent system. There are no solutions.
In this case, the intersection of any two planes is parallel to the third plane. {case **(8)**}

c If $a = 9$, the last row is all zeros, indicating infinitely many solutions.

Letting $z = t$ in row 2 gives
$y + \frac{7}{4}t = \frac{1}{4}$
$\therefore \quad y = \dfrac{1-7t}{4}$

Using row 1, $x + \left(\dfrac{1-7t}{4}\right) + 3t = 2$
$\therefore \quad x + \dfrac{5t}{4} = \dfrac{7}{4}$
$\therefore \quad x = \dfrac{7-5t}{4}$

We have infinitely many solutions of the form:
$x = \dfrac{7-5t}{4}, \quad y = \dfrac{1-7t}{4}, \quad z = t, \quad$ where $t \in \mathbb{R}$.
In this case we have three planes which meet in a line. {case **(7)**}

6 Solve the system of equations $\begin{cases} x - y + 3z = 1 \\ 2x - 3y - z = 3 \\ 3x - 5y - 5z = k \end{cases}$ where k is a real number.

State the geometrical meaning of the different solutions.

7 Solve the system of equations $\begin{cases} x - y + 2z = 1 \\ 3x + y + 2z = 0 \\ 5x + 3y + az = 2 \end{cases}$ where a is a real number.

State the geometrical meaning of the different solutions.

8 Consider the system of equations $\begin{cases} x + 2y + 3z = 4 \\ -x - 2y + 2z = 1 \\ 3x + 6y + kz = 5 \end{cases}$ where k is a real number.

a Reduce the system to row echelon form using row operations.

b Show that when $k \neq 2$, the system is inconsistent. For which of theses value(s) of k are two of the planes parallel? Give a geometric interpretation for the remaining values of k for which the system is inconsistent.

c Find the value(s) of k for which the system has infinitely many solutions. Interpret this result geometrically.

9 Reduce the system of equations: $\begin{cases} 2x + y - z = 3 \\ mx - 2y + z = 1 \\ x + 2y + mz = -1 \end{cases}$ to a form in which the solutions may be determined for all real values of m.

a Show that the system has no solutions for one value of m. Interpret this result geometrically.

b Show that there are infinitely many solutions for another value of m. Interpret this result.

c **i** For what values of m does the system have a unique solution? Interpret this result.

ii Show that the unique solution is $x = \dfrac{7}{m+5}, \quad y = \dfrac{3(m-2)}{m+5}, \quad z = \dfrac{-7}{m+5}$.

10 **a** Reduce this system to row echelon form: $\begin{cases} 2x - y + 3z = 4 \\ 2x + y + (a+3)z = 10 - a \\ 4x + 6y + (a^2+6)z = a^2 \end{cases}$

b Find the value(s) of a for which the system has no solutions. Interpret this result geometrically.

c Find the value(s) of a for which the system has infinitely many solutions, and find the form of these solutions. Interpret this result geometrically.

d Find the value(s) of a for which the system has a unique solution, and find the solution in the case where $a = 2$. Interpret this result geometrically.

11 Consider the system of equations $\begin{cases} x + 3y + 3z = a - 1 \\ 2x - y + z = 7 \\ 3x - 5y + az = 16 \end{cases}$ where a could take any real value.

a Reduce the system to row echelon form using row operations.

b Show that if $a = -1$ the system has infinitely many solutions, and find their form. Interpret this result geometrically.

c If $a \neq -1$, find the unique solution in terms of a. Interpret this result geometrically.

12 Find if and where the following planes meet:

P_1: $\mathbf{r}_1 = 2\mathbf{i} - \mathbf{j} + \lambda(3\mathbf{i} + \mathbf{k}) + \mu(\mathbf{i} + \mathbf{j} - \mathbf{k})$, $\lambda, \mu \in \mathbb{R}$

P_2: $\mathbf{r}_2 = 3\mathbf{i} - \mathbf{j} + 3\mathbf{k} + r(2\mathbf{i} - \mathbf{k}) + s(\mathbf{i} + \mathbf{j})$, $r, s \in \mathbb{R}$

P_3: $\mathbf{r}_3 = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + t\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} - u\begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}$, $t, u \in \mathbb{R}$

DISCUSSION

Click on the icon to find a printable summary for the equations of lines and planes.

SUMMARY



Discuss the similarities and differences between the equations.

REVIEW SET 13A

1 For the line that passes through $(-6, 3)$ with direction $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$, write down the corresponding:

a vector equation **b** parametric equations **c** Cartesian equation.

2 $(-3, m)$ lies on the line with vector equation $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 18 \\ -2 \end{pmatrix} + t\begin{pmatrix} -7 \\ 4 \end{pmatrix}$, $t \in \mathbb{R}$. Find m.

3 Consider A$(2, -1, 3)$ and B$(0, 1, -1)$.

a Find the vector equation of the line through A and B.

b Hence find the coordinates of C on (AB) which is 2 units from A.

4 **a** Consider two unit vectors **a** and **b**. Prove that the vector $\mathbf{a} + \mathbf{b}$ bisects the angle between vector **a** and vector **b**.

b Consider the points H$(9, 5, -5)$, J$(7, 3, -4)$, and K$(1, 0, 2)$.
Find the vector equation of the line L that passes through J and bisects $\widehat{\text{HJK}}$.

c Find the coordinates of the point where L meets (HK).

5 Triangle ABC is formed by three lines:

(AB) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} + t\begin{pmatrix} 1 \\ 3 \end{pmatrix}$, (BC) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 4 \end{pmatrix} + s\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, and

(AC) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + u\begin{pmatrix} 3 \\ 1 \end{pmatrix}$, where s, t, and u are scalars.

a Use vector methods to find the coordinates of A, B, and C.

b Find $|\overrightarrow{\text{AB}}|$, $|\overrightarrow{\text{BC}}|$, and $|\overrightarrow{\text{AC}}|$.

c Classify triangle ABC.

6 A yacht is sailing with constant speed $5\sqrt{10}$ km h^{-1} in the direction $-\mathbf{i} - 3\mathbf{j}$. Initially it is at point $(-6, 10)$. A beacon is at $(0, 0)$ at the centre of a tiny atoll. Distances are in kilometres.



a Find, in terms of **i** and **j**, the:

i initial position vector of the yacht

ii velocity vector of the yacht

iii position vector of the yacht after t hours, $t \geqslant 0$.

b Find the time when the yacht is closest to the beacon.

c If there is a reef of radius 8 km around the atoll, will the yacht hit the reef?

7 Submarine X23 is at $(2, 4)$. It fires a torpedo with velocity vector $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$ at exactly 2:17 pm.

Submarine Y18 is at $(11, 3)$. It fires a torpedo with velocity vector $\begin{pmatrix} -1 \\ a \end{pmatrix}$ at 2:19 pm to intercept the torpedo from X23. Distance units are kilometres, and time units are minutes.

a Find $x_1(t)$ and $y_1(t)$ for the torpedo fired from submarine X23.

b Find $x_2(t)$ and $y_2(t)$ for the torpedo fired from submarine Y18.

c At what time does the interception occur?

d What was the direction and speed of the intercepting torpedo?

e Find the acute angle between the paths of the torpedoes.

8 Find the coordinates of the foot of the perpendicular from Q$(-1, 2, 3)$ to the line $2 - x = y - 3 = -\frac{1}{2}z$.

9 **a** Find the Cartesian equation of the plane through A$(-1, 0, 2)$, B$(0, -1, 1)$, and C$(1, 2, -1)$.

b Find the equation of the line, in parametric form, which passes through the origin and is normal to the plane in **a**.

c Find the point where the line in **b** intersects the plane in **a**.

10 Consider the lines L_1: $\dfrac{x-8}{3} = \dfrac{y+9}{-16} = \dfrac{z-10}{7}$

and L_2: $x = 15+3t,\ y = 29+8t,\ z = 5-5t,\ t \in \mathbb{R}$

a Show that the lines are skew. **b** Find the acute angle between them.

c Find the shortest distance between them.

11 Suppose A is $(3,\ 2,\ -1)$ and B is $(-1,\ 2,\ 4)$.

a Write down a vector equation of the line through A and B.

b Find the Cartesian equation of the plane through B with normal $\overrightarrow{AB}$.

c Find *two* points on (AB) which are $2\sqrt{41}$ units from A.

12 For $C(-3,\ 2,\ -1)$ and $D(0,\ 1,\ -4)$, find the coordinates of the point where the line passing through C and D meets the plane with equation $2x - y + z = 3$.

13 **a** Find if possible the point where the line through $L(1,\ 0,\ 1)$ and $M(-1,\ 2,\ -1)$ meets the plane with equation $x - 2y - 3z = 14$.

b Find the shortest distance from L to the plane.

14 Given the point $A(-1,\ 3,\ 2)$, the plane $2x - y + 2z = 8$, and the line defined by $x = 7-2t,\ y = -6+t,\ z = 1+5t,\ t \in \mathbb{R}$, find:

a the distance from A to the plane

b the coordinates of the point on the plane nearest to A

c the shortest distance from A to the line.

15 Given $A(-1,\ 2,\ 3)$, $B(1,\ 0,\ -1)$, and $C(1,\ 3,\ 0)$, find:

a the normal vector to the plane containing A, B, and C

b D, the fourth vertex of parallelogram ACBD

c the area of parallelogram ACBD

d the coordinates of the foot of the perpendicular from C to the line (AB).

16 Lines L_1 and L_2 are defined by

$$L_1:\ \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix} + s\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} \quad \text{and} \quad L_2:\ \mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} + t\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}, \quad \text{where } s, t \in \mathbb{R}.$$

a Find the coordinates of A, the point of intersection of the lines.

b Show that the point $B(0,\ -3,\ 2)$ lies on the line L_2.

c Find the vector equation of the line (BC) given that $C(3,\ -2,\ -2)$ lies on L_1.

d Find the Cartesian equation of the plane containing A, B, and C.

e Find the area of triangle ABC.

f Show that the point $D(9,\ -4,\ 2)$ lies on the normal to the plane passing through C.

17 Consider $A(-1,\ 2,\ 3)$, $B(2,\ 0,\ -1)$, and $C(-3,\ 2,\ -4)$.

a Find the Cartesian equation of the plane defined by A, B, and C.

b Find the measure of $C\widehat{A}B$.

c $D(r,\ 1,\ -r)$ is a point such that $B\widehat{D}C$ is a right angle. Find r.

18 Suppose P_1 is the plane $2x - y - 2z = 9$ and P_2 is the plane $x + y + 2z = 1$.
L is the line with parametric equations $x = t,\ y = 2t - 1,\ z = 3 - t,\ t \in \mathbb{R}$.
Find the acute angle between:

a L and P_1 **b** P_1 and P_2.

19 The three planes P_1, P_2, and P_3 are mutually perpendicular. P_1 has equation $ax + 3y - z = 2$, P_2 has equation $bx + 2z = 5$, and P_3 has equation $cx - 5y - 3z = 20$, where a, b, and c are positive constants.

a Find a, b, and c. **b** Find the point of intersection of the planes.

c Find the equation of L, the line of intersection of P_1 and P_2.

d Show that $\mathrm{P}(2,\ 2,\ -6)$ lies on P_3.

e Find the equation of P_4, the plane containing L and P.

f Find the angle between P_1 and P_4.

20 Suppose $\mathbf{p} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$.

a Find $\mathbf{p} \times \mathbf{q}$.

b Find m if $\mathbf{p} \times \mathbf{q}$ is perpendicular to the line L with equation $\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ m \end{pmatrix}$, $\lambda \in \mathbb{R}$.

c Hence find the Cartesian equation of the plane P containing L which is perpendicular to $\mathbf{p} \times \mathbf{q}$.

d Find t if the point $\mathrm{A}(4,\ t,\ 2)$ lies on the plane P.

e Suppose B is the point $(6,\ -3,\ 5)$. For the value of t found in **d**, find the angle between (AB) and the plane P.

21 Use row operations to solve the system $\begin{cases} x - y + z = 5 \\ 2x + y - z = -1 \\ 7x + 2y + kz = -k \end{cases}$, $k \in \mathbb{R}$. Give geometric interpretations of your results.

22 **a** Show that the plane $2x + y + z = 5$ contains the line
L_1: $x = -2t + 2,\quad y = t,\quad z = 3t + 1,\quad t \in \mathbb{R}$.

b For what values of k does the plane $x + ky + z = 3$ contain L_1?

c *Hence* find the values of p and q for which the system $\begin{cases} 2x + y + z = 5 \\ x - y + z = 3 \\ -2x + py + 2z = q \end{cases}$ has an infinite number of solutions. Clearly explain your reasoning.

REVIEW SET 13B

1 Line L has equation $\mathbf{r} = \begin{pmatrix} 3 \\ -3 \end{pmatrix} + t \begin{pmatrix} 2 \\ 5 \end{pmatrix},\ t \in \mathbb{R}$.

a Locate the point on the line corresponding to $t = 1$.

b Explain why the direction of the line could also be described by $\begin{pmatrix} 4 \\ 10 \end{pmatrix}$.

c Use your answers to **a** and **b** to write an alternative vector equation for line L.

2 Write down **i** a vector equation **ii** parametric equations for the line passing through:

a $(2, -3, 1)$ with direction $\begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}$ **b** $(-1, 6, 3)$ and $(5, -2, 0)$.

3 Find the angle between line L_1 passing through $(0, 3)$ and $(5, -2)$, and line L_2 passing through $(-2, 4)$ and $(-6, 7)$.

4 Trapezium KLMN is formed by the following lines:

(KL) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 19 \end{pmatrix} + p\begin{pmatrix} 5 \\ -2 \end{pmatrix}$, (ML) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 33 \\ -5 \end{pmatrix} + q\begin{pmatrix} -11 \\ 16 \end{pmatrix}$,

(NK) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix} + r\begin{pmatrix} 4 \\ 10 \end{pmatrix}$, and (MN) is $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 43 \\ -9 \end{pmatrix} + s\begin{pmatrix} -5 \\ 2 \end{pmatrix}$,

where p, q, r, and s are scalars.

a Which two lines are parallel? Explain your answer.

b Which lines are perpendicular? Explain your answer.

c Use vector methods to find the coordinates of K, L, M, and N.

d Calculate the area of trapezium KLMN.

5 Find the angle between the lines:

L_1: $x = 1 - 4t$, $y = 3t$ and L_2: $x = 2 + 5s$, $y = 5 - 12s$, where $t, s \in \mathbb{R}$.

6 Consider $\mathrm{A}(3, -1, 1)$ and $\mathrm{B}(0, 2, -2)$.

a Find $|\overrightarrow{\mathrm{AB}}|$.

b Show that the line passing through A and B can be described by $\mathbf{r} = 2\mathbf{j} - 2\mathbf{k} + \lambda(-\mathbf{i} + \mathbf{j} - \mathbf{k})$ where $\lambda \in \mathbb{R}$.

c Find the angle between (AB) and the line with equation $\mathbf{r} = t(\mathbf{i} + \mathbf{j} + \mathbf{k})$, $t \in \mathbb{R}$.

7 Find the velocity vector of an object moving in the direction $3\mathbf{i} - \mathbf{j}$ with speed 20 km h^{-1}.

8 A moving particle has coordinates $\mathrm{P}(x(t), y(t), z(t))$ where $x(t) = -4 + 8t$, $y(t) = 3 + 6t$, and $z(t) = 1 - 2t$. The distance units are metres, and $t \geqslant 0$ is the time in seconds. Find the:

a initial position of the particle **b** position of the particle after 4 seconds

c particle's velocity vector **d** speed of the particle.

9 The vector equation $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -10 \\ 5 \\ 12 \end{pmatrix} + t\begin{pmatrix} 6 \\ 14 \\ -0.4 \end{pmatrix}$ represents the path of a person zip-lining through a forest. t is measured in seconds, $t \geqslant 0$. Distances are measured in metres. Find the:

a initial position **b** velocity vector

c speed of the zip-liner

d time taken to reach the end of the line at $z = 0$

e location of the endpoint of the line

f position of the person when he is closest to his friend watching from $(52, 144, 3)$.



10 Boat A's position is given by $x_A(t) = 3 - t$, $y_A(t) = 2t - 4$, and boat B's position is given by $x_B(t) = 4 - 3t$, $y_B(t) = 3 - 2t$. The distance units are kilometres, and the time units are hours.



a Find the initial position of each boat.

b Find the velocity vector of each boat.

c Find the acute angle between the paths of the boats.

d At what time are the boats closest to each other?

11 Classify the following line pairs as either parallel, intersecting, or skew. Where possible, state the point of intersection and find the measure of the acute angle between them.

a $x = 2 + t$, $y = -1 + 2t$, $z = 3 - t$ and $x = -8 + 4s$, $y = s$, $z = 7 - 2s$

b $x = 3 + t$, $y = 5 - 2t$, $z = -1 + 3t$ and $x = 2 - s$, $y = 1 + 3s$, $z = 4 + s$

c $\dfrac{x-3}{2} = y - 4 = \dfrac{z+1}{-2}$ and $x = -1 + 3t$, $y = 2 + 2t$, $z = 3 - t$

12 **a** Find the Cartesian equation of the plane through A$(-1,\ 2,\ 3)$, B$(1,\ 0,\ -1)$, and C$(0,\ -1,\ 5)$.

b If X is $(3,\ 2,\ 4)$, find the angle that (AX) makes with this plane.

13 Find the coordinates of N, the foot of the normal from A$(1,\ 5,\ 3)$ to the plane with equation $\mathbf{r} = -5\mathbf{i} - \mathbf{j} - 5\mathbf{k} + s(3\mathbf{i} + \mathbf{j} - \mathbf{k}) + t(-\mathbf{i} + 2\mathbf{j} + \mathbf{k})$, $s, t \in \mathbb{R}$.

14 How far is X$(-1,\ 1,\ 3)$ from the plane $x - 2y - 2z = 8$?

15 The equations of two lines are: L_1: $x = 3t - 4$, $y = t + 2$, $z = 2t - 1$, $t \in \mathbb{R}$

and L_2: $x = \dfrac{y-5}{2} = \dfrac{-z-1}{2}$.

a Find the point of intersection of L_1 and the plane $2x + y - z = 2$.

b Find the point of intersection of L_1 and L_2.

c Find the equation of the plane that contains L_1 and L_2.

16 Show that the line $x - 1 = \dfrac{y+2}{2} = \dfrac{z-3}{4}$ is parallel to the plane $6x + 7y - 5z = 8$ and find the distance between them.

17 P$(2,\ 0,\ 1)$, Q$(3,\ 4,\ -2)$, and R$(-1,\ 3,\ 2)$ are three points in space. Find:

a $\overrightarrow{PQ}$, $|\overrightarrow{PQ}|$, and $\overrightarrow{QR}$ **b** the parametric equations of (PQ)

c a vector equation of the plane through P, Q, and R.

18 Suppose π_1 is the plane $x - 2y + 4z = 21$, π_2 is the plane $5x - y + 3z = 2$, and L is the line $\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} + t \begin{pmatrix} 6 \\ 1 \\ -1 \end{pmatrix}$, $t \in \mathbb{R}$.

a Show that π_1 contains L. **b** Find the point at which L intersects π_2.

c Find the angle between:

i L and π_2 **ii** π_1 and π_2.

19 Line 1 has parametric equation $x = s,\ y = 2 - s,\ z = -2 + s,\ s \in \mathbb{R}$.

Line 2 has vector equation $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 6 \end{pmatrix} + t\begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix},\ t \in \mathbb{R}$.

a Show that lines 1 and 2 are skew.

b Find the shortest distance between them.

c Find the coordinates of the points where the common perpendicular meets the lines 1 and 2.

20 Find the angle between the plane $2x + 2y - z = 3$ and the line $x = t - 1,\ y = -2t + 4,\ z = -t + 3,\ t \in \mathbb{R}$.

21 Consider the following system of linear equations in which p and q are constants:

$$\begin{cases} x - 2y + 3z = 1 \\ x + py + 2z = 0 \\ -2x + p^2y - 4z = q \end{cases}$$

a Show that this system can be reduced to $\left(\begin{array}{ccc|c} 1 & -2 & 3 & 1 \\ 0 & p+2 & -1 & -1 \\ 0 & 0 & p & p+q \end{array}\right)$.

b What values can p and q take if the system has:

i a unique solution **ii** no solutions **iii** infinite solutions?

Explain the geometric significance of each case.

c Specify the infinite solutions in parametric form.

22 Three planes have the equations given below:

Plane A: $x + 3y + 2z = 5$

Plane B: $2x + y + 9z = 20$

Plane C: $x - y + 6z = 8$

a Show that plane A and plane B intersect in a line L_1.

b Show that plane B and plane C intersect in a line L_2.

c Show that plane A and plane C intersect in a line L_3.

d Show that L_1, L_2, and L_3 are parallel but not coincident.

e What does this mean geometrically?

23 Consider the system $\begin{cases} x - 3y + 2z = -5 \\ 3x + y + (2-k)z = 10 \\ -2x + 6y + kz = 5 \end{cases}$ where k can take any real value.

a Reduce the system to row echelon form.

b For what value of k does the system have no solutions? Interpret this result geometrically.

c **i** For what value(s) of k does the system have a unique solution?

ii Find the unique solution in terms of k, and interpret the result geometrically.

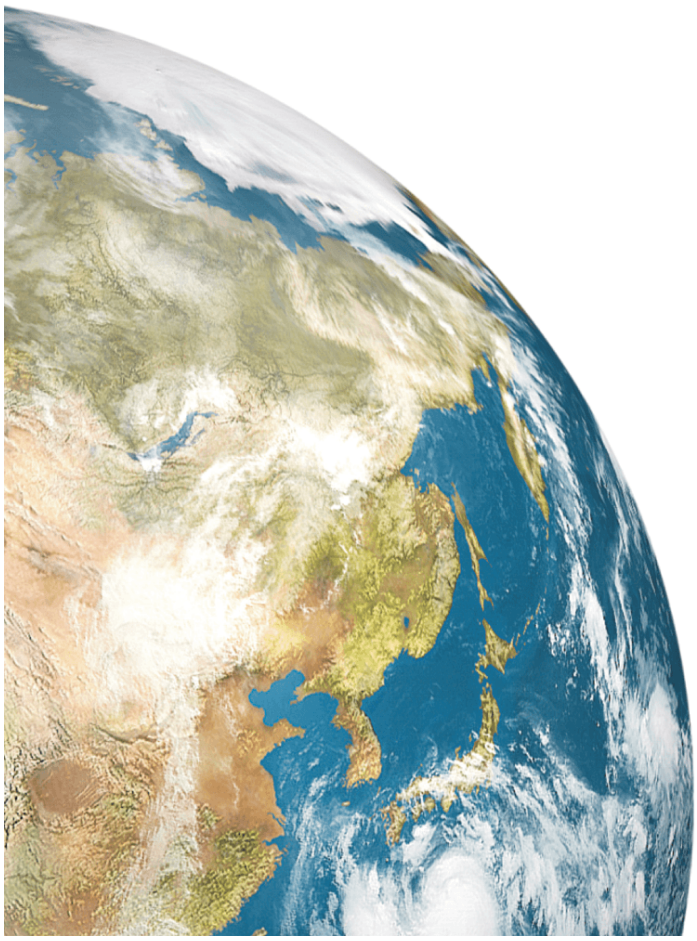
iii Find the unique solution when $k = 1$.

Chapter 14

Complex numbers

Contents:

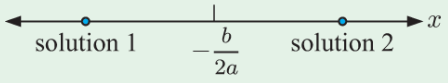
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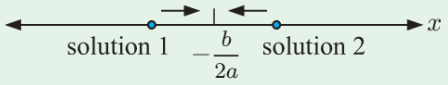
OPENING PROBLEM

Consider a quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$ with real coefficients.

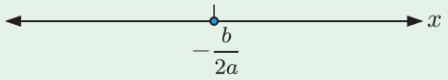
If $\Delta > 0$, there are two distinct real solutions with average value $-\frac{b}{2a}$.



If the value of c is changed so that Δ decreases towards zero, the solutions get closer together.



When $\Delta = 0$, the solutions coincide.



Things to think about:

a Suppose Δ is decreased further so there are two *complex* solutions.

- **i** What is the real component of these solutions?
- **ii** How can we *illustrate* the solutions so we can see both the real and imaginary parts?

b How can we use a *vector* to represent a complex number?

c
- **i** Can you illustrate *all* of the solutions to $x^2 = 1$, $x^3 = 1$, $x^4 = 1$, $x^5 = 1$, and $x^6 = 1$?
- **ii** What do you notice about the *magnitude* of the vectors which represent these solutions?

d
- **i** Can you illustrate *all* of the solutions to $x^3 = 1$, $x^3 = 8$, $x^3 = 27$, $x^3 = 64$, and $x^3 = 125$?
- **ii** What do you notice about the *directions* of the vectors which represent these solutions?

A THE COMPLEX PLANE

We have seen that a complex number can be written in Cartesian form as $z = a + bi$ where $a = \mathcal{Re}(z)$ and $b = \mathcal{Im}(z)$ are both real numbers.

There is a one-to-one relationship between any complex number $a + bi$ and the point (a, b) in the Cartesian plane.

We can therefore plot any complex number on a plane as a unique ordered number pair. We refer to the plane as the **complex plane** or the **Argand plane**.

On the complex plane, the x-axis is called the **real axis** and the y-axis is called the **imaginary axis**.

- All **real** numbers with $b = 0$ lie on the real axis.
- All **purely imaginary** numbers with $a = 0$ lie on the imaginary axis.
- The origin $(0, 0)$ lies on both axes. It corresponds to $z = 0$, a real number.
- Complex numbers that are neither real nor purely imaginary (a and b are both not 0) lie in one of the four quadrants.

